

Lecture 1.1: GIS Overview and Geospatial Data Types

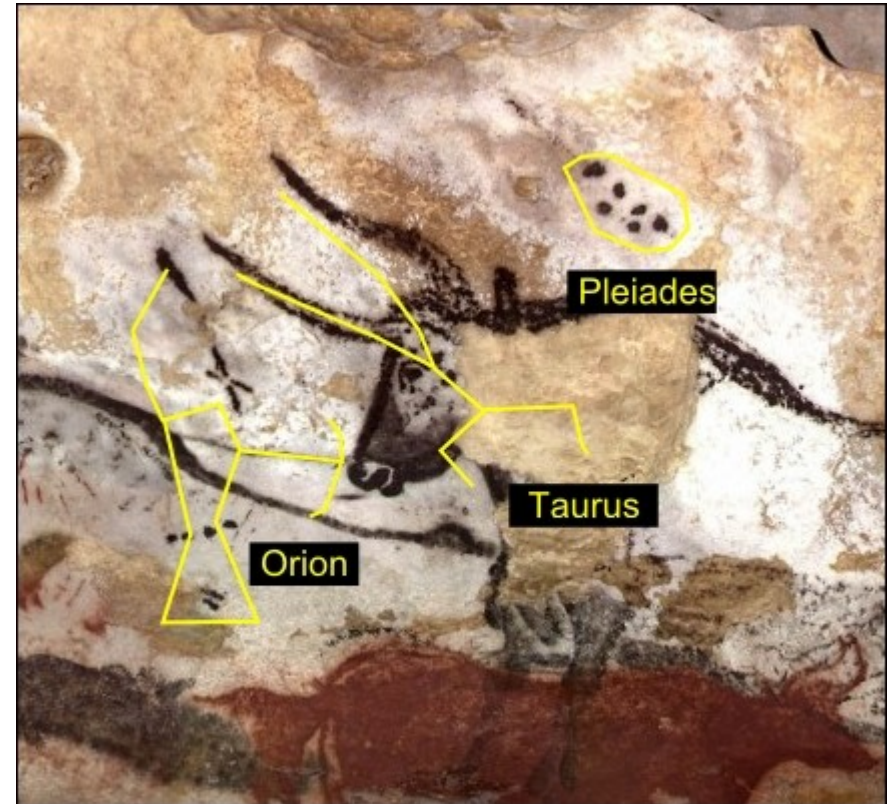
CSE620B: Remote Sensing & Computer Vision

Dr. John C. Femiani

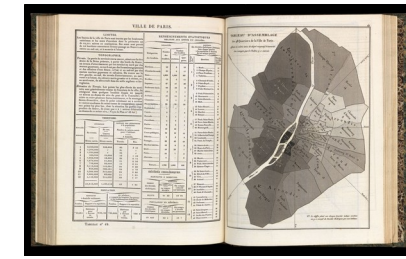
Miami University

Early History of Geospatial Analysis

- Geospatial analysis has roots tracing back 15,000 years.
- Example: The Lascaux cave in southwestern France, where Paleolithic artists painted animals and star maps.
- These early maps may have been used for religious ceremonies or tracking migration patterns of prey.
- Represents one of the earliest examples of abstract modeling and correlating spatial-temporal features.



Cholera Outbreaks



- **19th Century Geographic Analysis**

The cholera outbreaks in Europe between 1830 and 1860 prompted the use of geographic analysis in epidemiology.

- **Charles Picquet (1832)**

- Used half-toned shades of gray to represent cholera deaths per thousand citizens in Paris.

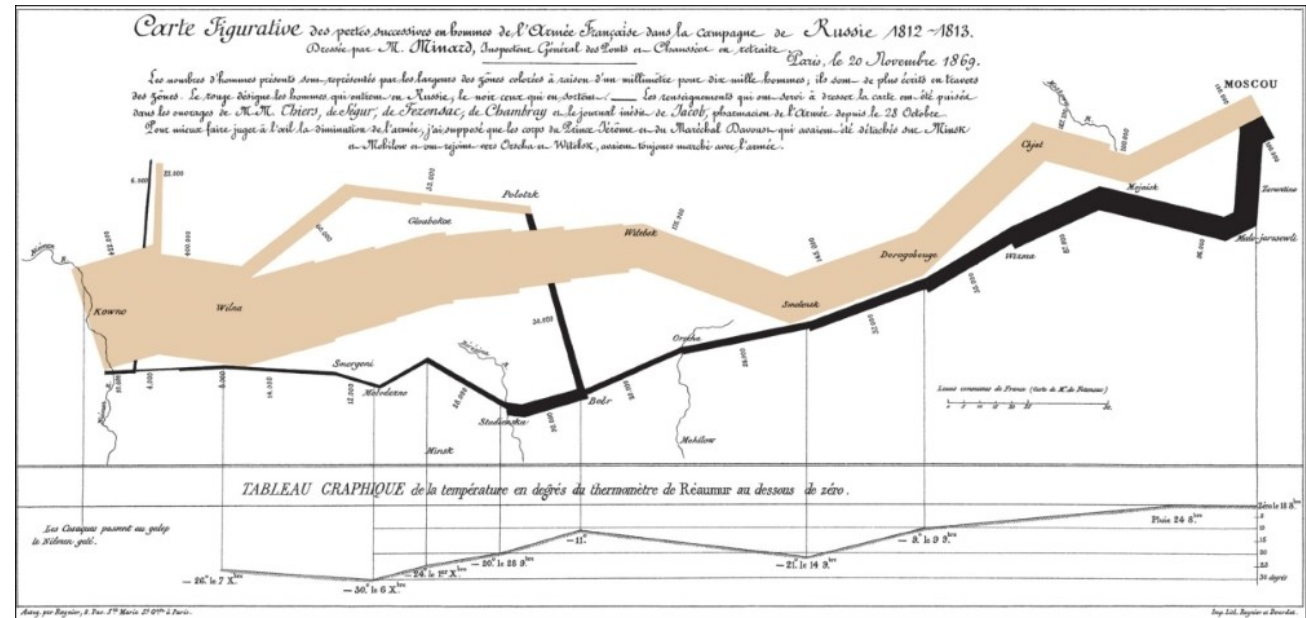
- **Dr. John Snow (1854)**

- Expanded on Picquet's methods by tracking cholera deaths in London.
- Mapped cholera deaths to specific locations, enabling him to trace the outbreak to a contaminated water pump.
- This pioneering work in spatial analysis helped prevent further deaths by identifying the source of contamination.



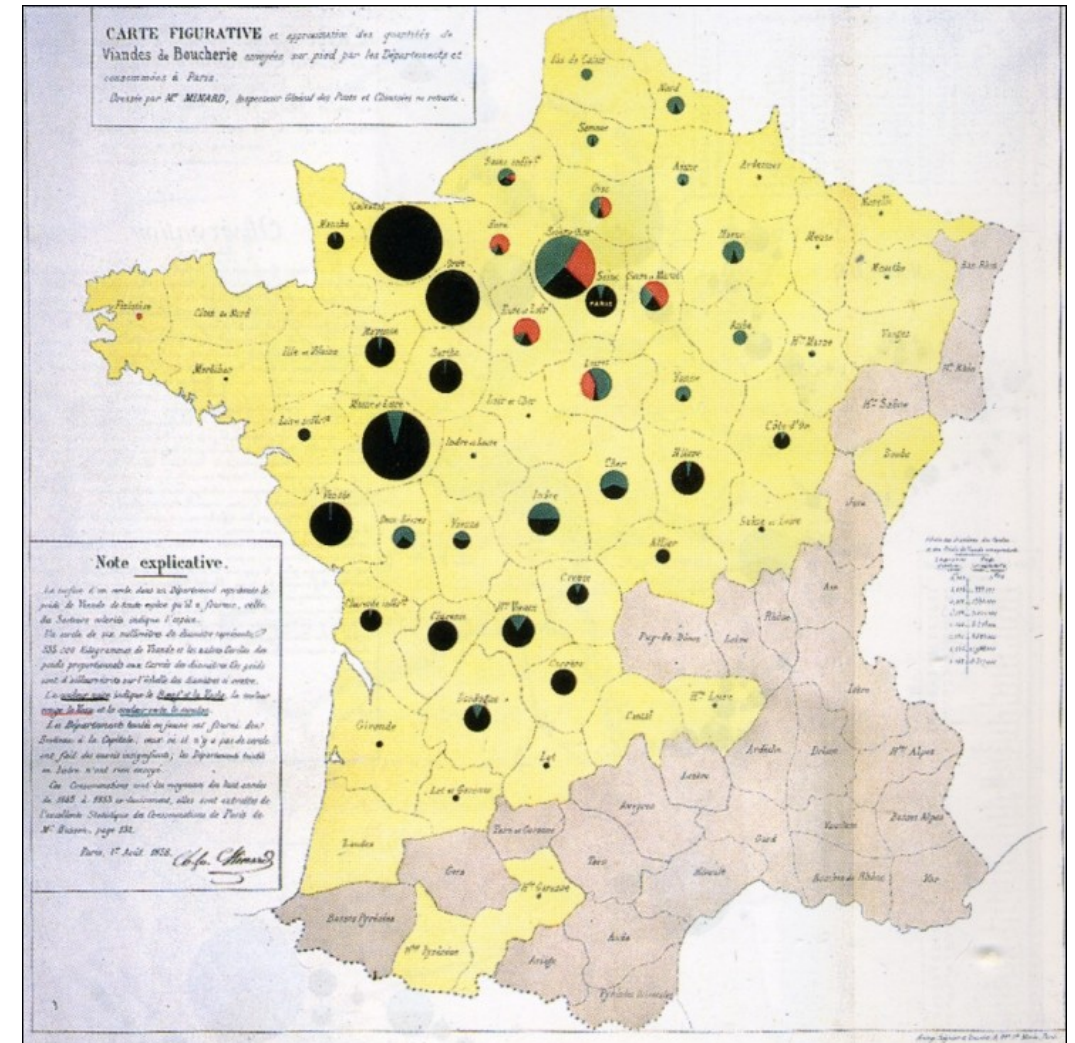
Geospatial Analysis – Napoleon's War in Russia

- Charles Minard created sophisticated geospatial infographics in the 1800s.
- His 1869 masterpiece depicts Napoleon's 1812-1813 Russian campaign.
- The brown swath shows soldiers entering Russia; the black swath shows those returning.
- The width of the swaths represents army size (1 mm per 10,000 men).
- A chart at the bottom shows the freezing temperatures during the army's retreat.



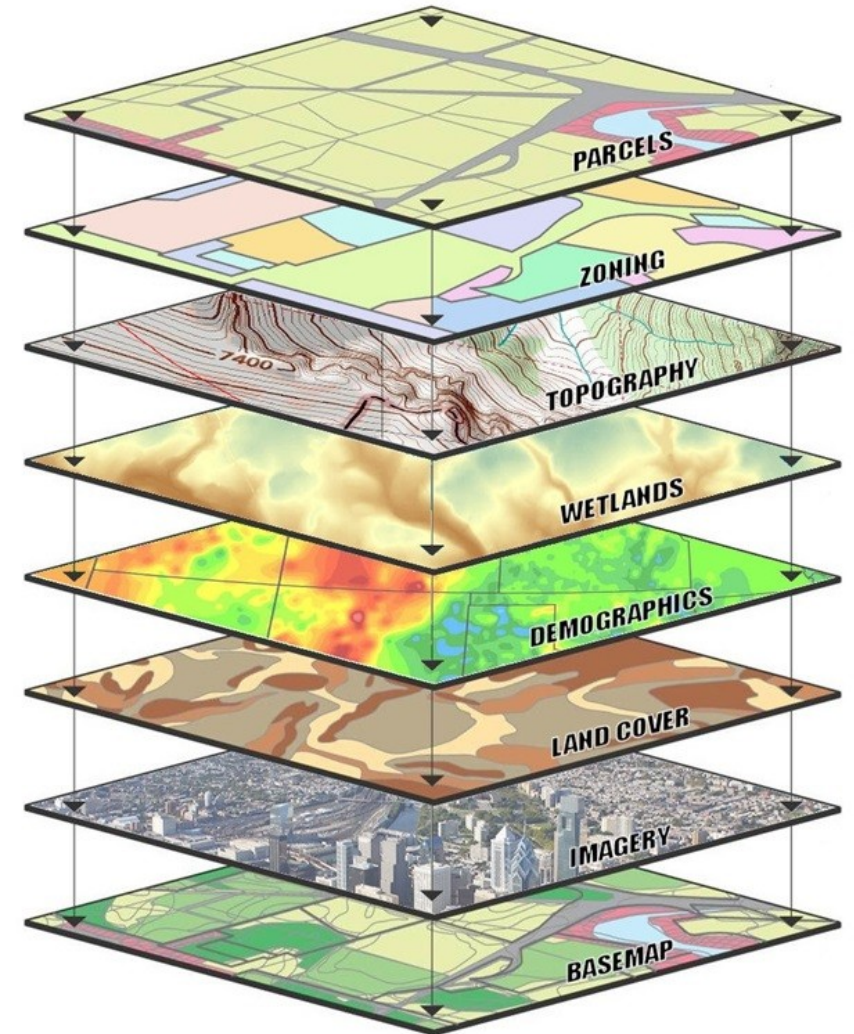
Geospatial Analysis -- Agriculture

- Charles Minard also used geospatial analysis for agriculture.
- His map shows the number of cattle sent to Paris from different regions of France.
- Pie charts represent the volume and variety of cattle shipped from each region.



Map Layers in the Early 1900s

- The early 1900s saw the development of map layers, a crucial feature of modern geospatial analysis.
- Cartographers used glass plates to separate elements like vegetation, roads, and elevation contours.
- This technique made it easier to correct mistakes and streamline the map-making process.
- The layering concept became a powerful tool for analysis with the advent of computers.



The Birth of Geographic Information Systems (GIS)

- Computer mapping evolved in the 1960s.
- GIS originated with the Canadian Geographic Information System (CGIS).
- Dr. Roger Tomlinson led the development of CGIS with IBM.
- CGIS tracked Canada's natural resources using layers for land cover.
- Dr. Tomlinson is known as the father of GIS.
- In this class, we will focus on open-source tools like QGIS.



[Roger Tomlinson \(c. 1965\)](#)



The Beginnings of Remote Sensing

- Remote sensing collects data without physical contact.
- In geospatial analysis, it focuses on Earth observation.
- Remote sensing reduced the cost of data collection for GIS.
- It enables automated feature extraction for GIS.
- First aerial photo: 1858 by Gaspard-Félix Tournachon (Nadar) over Paris.





FIGURE 1.1 Early aerial photography by the U.S. Navy, 1914. This photograph illustrates the difficulties encountered in early efforts to match the camera with the airplane. At that time, neither device was well suited for use with the other. From U.S. Navy, National Archives and Records Administration, ARC 295605.



FIGURE 1.2 Early aerial photography, World War I (ca. 1917–1918). By the time of World War I, attempts to match the camera and the airplane had progressed only to a modest extent, as illustrated by this example. This photographer, using a Graflex camera (here, likely posed for a publicity photograph), designed by the U.S. Army Signal Corps, aims the camera over the edge of the fuselage. The wooden camera body, with leather jacket, and focal plane shutter, required use of glass plates (likely 4 × 5 in.). Photographers operated within the cold slip-stream, manually changing plates in flight. From U.S. Army.



FIGURE 1.3 Aircraft showing later designs (relative to those in **Figures 1.1 and 1.2**) for both the biplane and the aerial camera. Here, the aerial camera has a lateral port (the square shape near the photographer, designed for oblique photography to allow the photographer to aim the camera through the port from within the fuselage, to avoid the disadvantages of leaning over edges of the cockpit. The photographer wears a chest-mounted microphone for communication with the pilot, shown here holding a supply of extra plates for the camera. From U.S. National Archives and Records Administration, Still Pictures, E-4156.

Institutionalization of Aerial Photography (1939–1945)

- USGS Adoption and Expansion
- Economic Pressures During Great Depression
- Environmental Monitoring Needs
- WWII Military Applications
- Transition to Civilian Use Post-War



FIGURE 1.5 Progress in applications of aerial photography, 1919–1939. During the interval between World War I and World War II (1919–1939), integration of the camera and the airplane progressed, as did institutionalization of aerial photography in government and industry. By June 1943 (the date of this photograph), progress on both fronts was obvious. Here, an employee of the U.S. Geological Survey (USGS) uses a specialized instrument, the *Oblique Sketchmaster*, to match detail on an aerial photograph to an accurate map. By the time of this photograph, aerial photography formed an integral component of USGS operations. From USGS and U.S. Library of Congress, digital ID fsa.8d38549.



FIGURE 1.4 Aerial image acquired by the K-2 Eastman mapping camera, illustrating post-World War I aerial cameras with fuselage-mounted aerial cameras designed for vertical aerial photography. Here, the white plate positioned at the top of the image frame provides real-time records of altitude, date, time of day, and orientation of the lens. From Lee (1922).



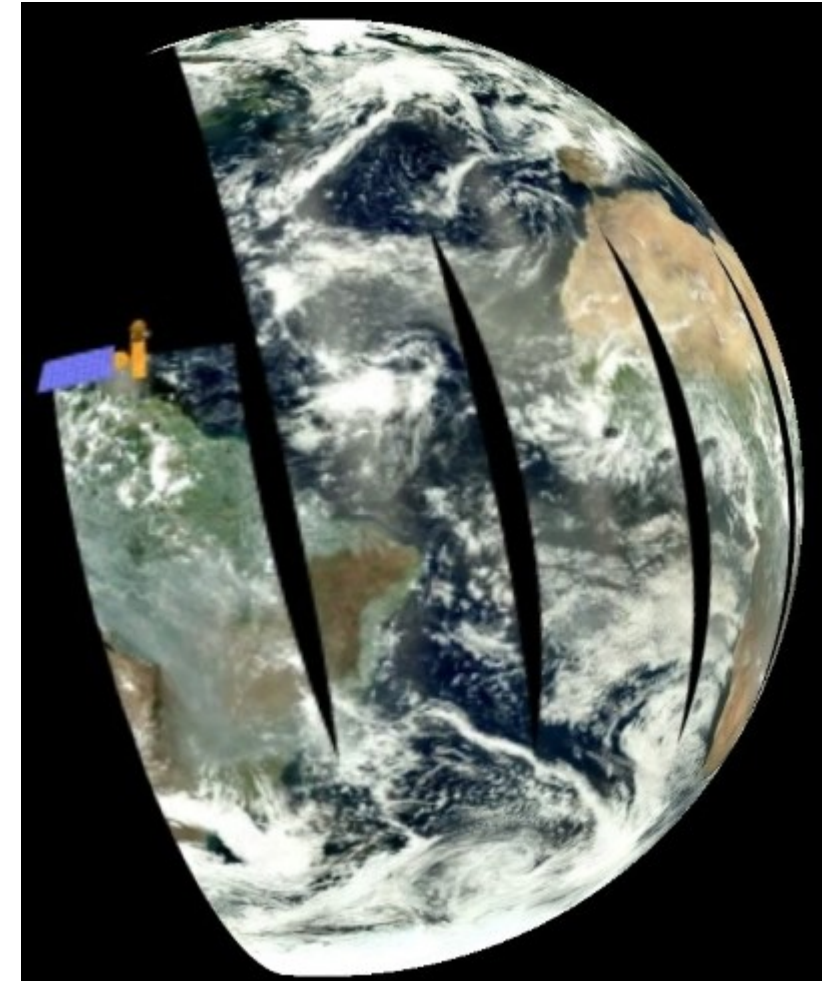
FIGURE 1.8 A *Twinplex* stereoscopic plotting instrument designed to prepare topographic maps to derive accurate elevation data from aerial photography (1957). During much of the twentieth century, photogrammetric analyses depended on optical-mechanical instruments such as shown here, designed to extract information by controlling the physical orientation of the photograph and optical projection of the image. By the end of the century, such processes were conducted in the digital domain using electronic instruments. From Photographic Library, USGS. Photograph by E. F. Patterson, no. 223.



FIGURE 1.12 A USGS cartographic technician using an airbrush to depict relief, as interpreted from aerial photographs, 1961. Within a few decades, computer cartography and GIS would routinely create this effect by applying hill-shading algorithms to create digital elevation models. From Photographic Library, USGS. Photograph by E. F. Patterson, no. 1024.

The Cold War and Early Satellite Remote Sensing

- **Cold War Era:** Aerial photography became prominent, especially with the U-2 spy plane.
- **Sputnik (1957):** The Soviet Union launched the first satellite, triggering the space race.
- **CORONA Program:** The U.S. developed the first satellite imagery system, using film canisters ejected from space and caught by planes.
<http://heroicrelics.org/info/corona>
- **Landsat-1 (1972):** NASA launched the first civilian remote sensing satellite, capturing global multispectral data.



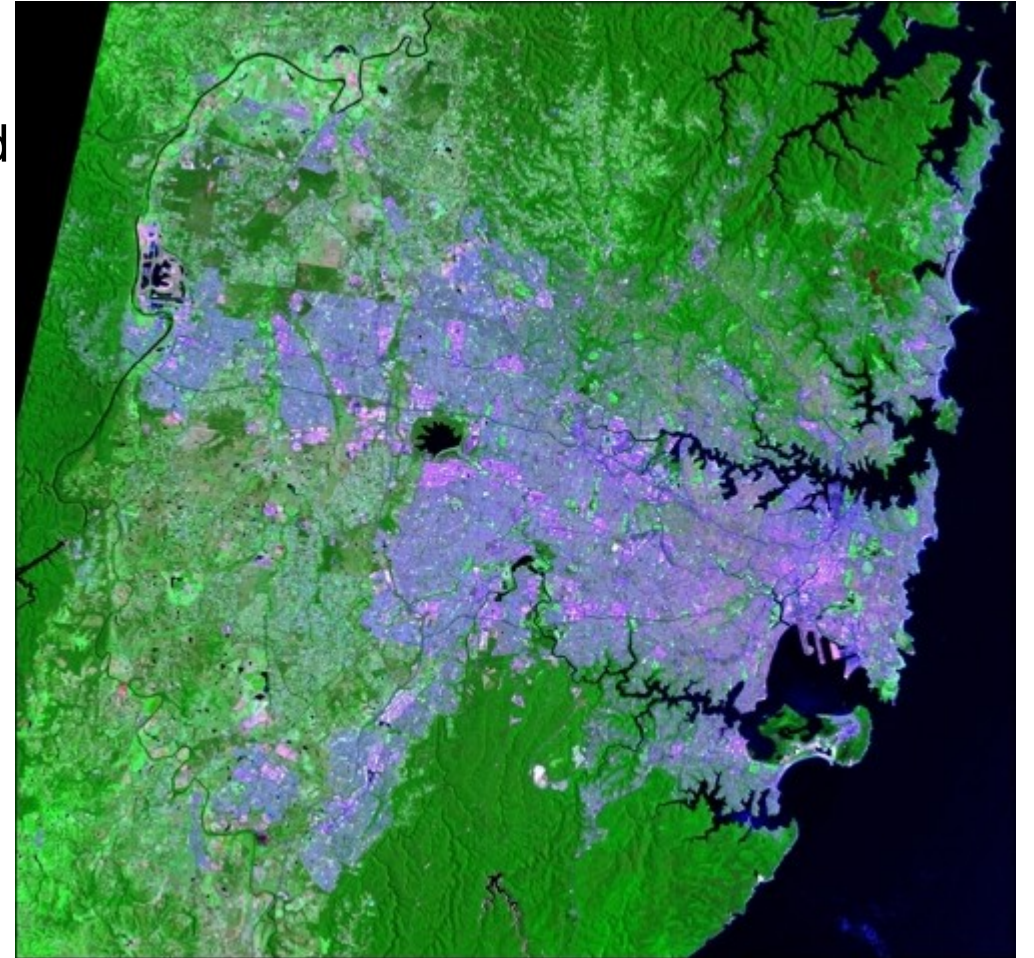
Earth with satellite swaths, showing how the satellite captures data tiles as it orbits.



FIGURE 1.14 NASA-Goddard staff working with Landsat/TIRS thermal imaging, March 13, 2012. Landsat's thermal instruments can record landscape temperatures from satellite altitudes to monitor water usage and water resources. Data from these instruments have provided reliable information documenting the availability of water resources for southwestern agriculture. From NASA-Goddard.

Landsat Missions

- The Landsat program started with **Landsat 1** and continued with six more missions.
- NOAA and EOSAT managed the satellites at different points, but control returned to the government with **Landsat 7**.
- **Landsat 6** failed to reach orbit due to a technical issue.
- **Landsat 8**, launched in 2013 as part of the **Landsat Data Continuity Mission (LDCM)**, is managed by NASA and the **USGS**.



Digital Elevation Models (DEMs)

- DEMs are 3D representations of Earth's terrain.
- The first usable DEM was created by the **SPOT-1 satellite** in 1986.
- **SRTM (2000)**: Collected elevation data over 80% of Earth's surface.
- **ASTER (2009)**: Captured 99% of the Earth's surface, with some data issues.
- **WorldDEM (2014)**: Launched by Germany, provides global DEM with high accuracy.



The Role of UAVs in Modern Remote Sensing

- Flexibility and High-Resolution Imaging
- Key Components: Sensors, Control, Data Links
- UAVs in Environmental Monitoring
- Complementary to Satellite Data



FIGURE 1.16 A systems analyst examining data and video transmitted by a U.S. Navy UAV to acquire intelligence, conduct surveillance, and collect reconnaissance data. From U.S. Marine Corps. Photograph by Cpl. Michael P. Snody.



Introduction to GIS

- **Key Concepts**

- Location, Distance, Projections, Coordinate Systems

- **Data Formats**

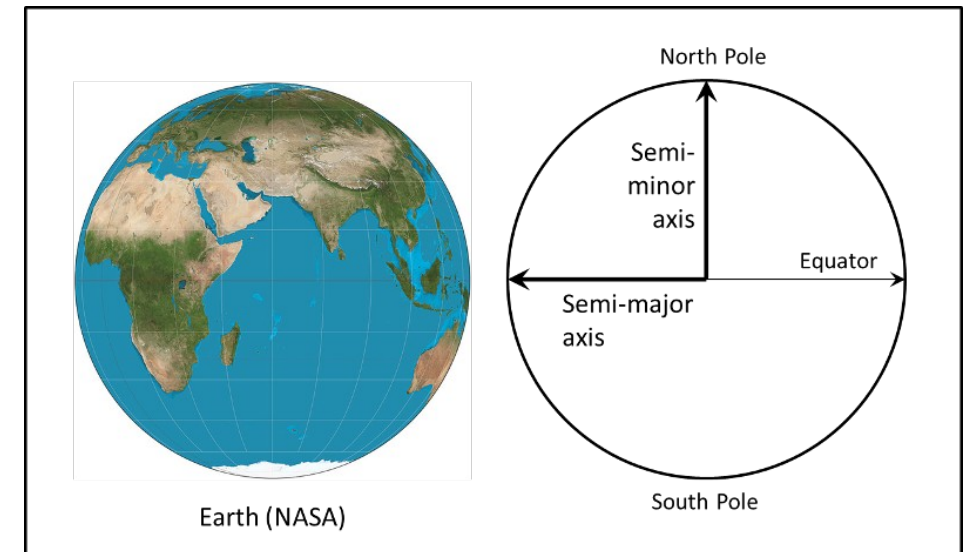
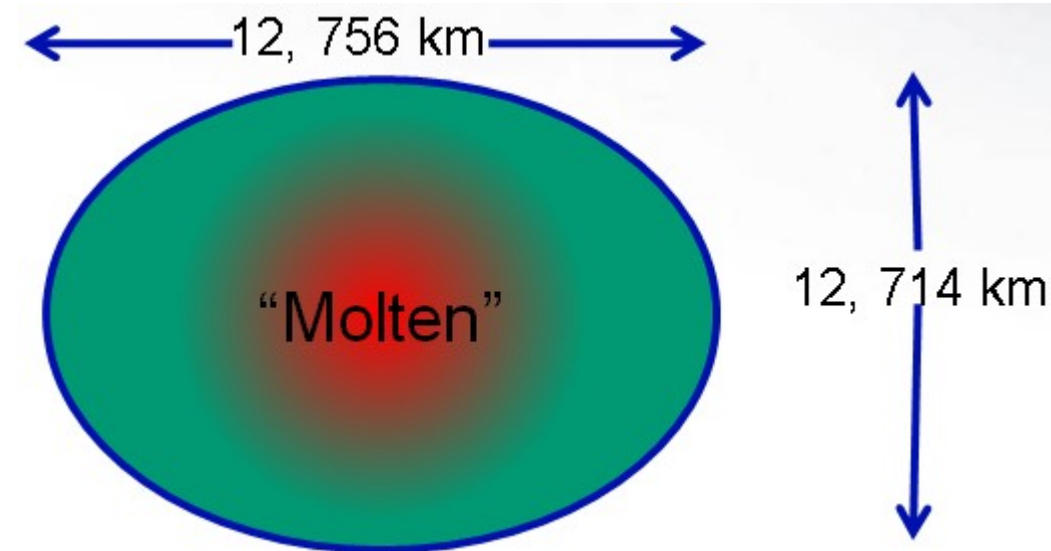
- Shapefiles, GeoJSON, Raster (GeoTIFF)

- **Processes**

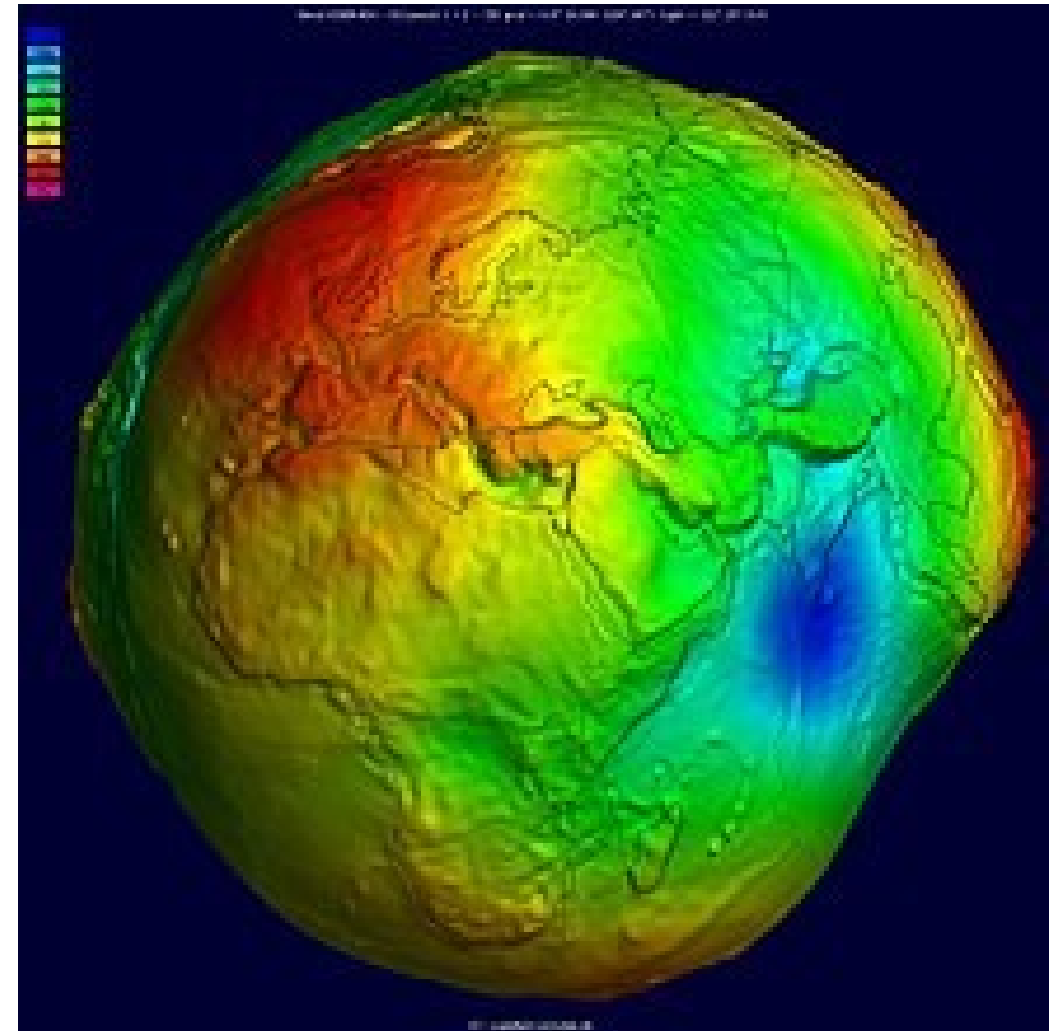
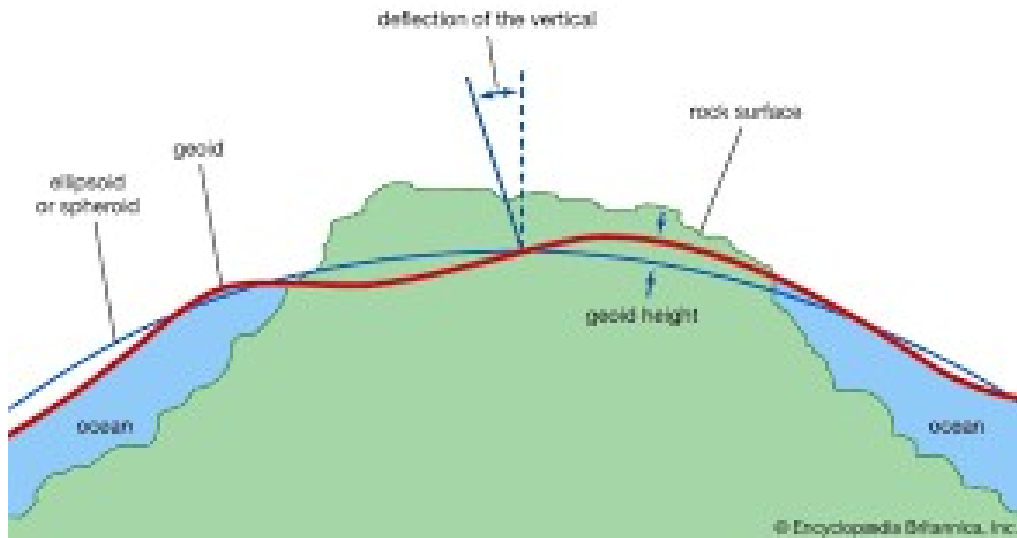
- Data Collection, Manipulation, Visualization

Core GIS Concepts

- **Earth's Shape**
 - Oblate Spheroid, not a perfect sphere
- **Geospatial Representation**
 - Points, Lines, Areas on Earth's surface
- **Mathematical Models**
 - Complexities in projecting Earth's surface



Actually worse than that!



Understanding Location in GIS

- **Location Representation**

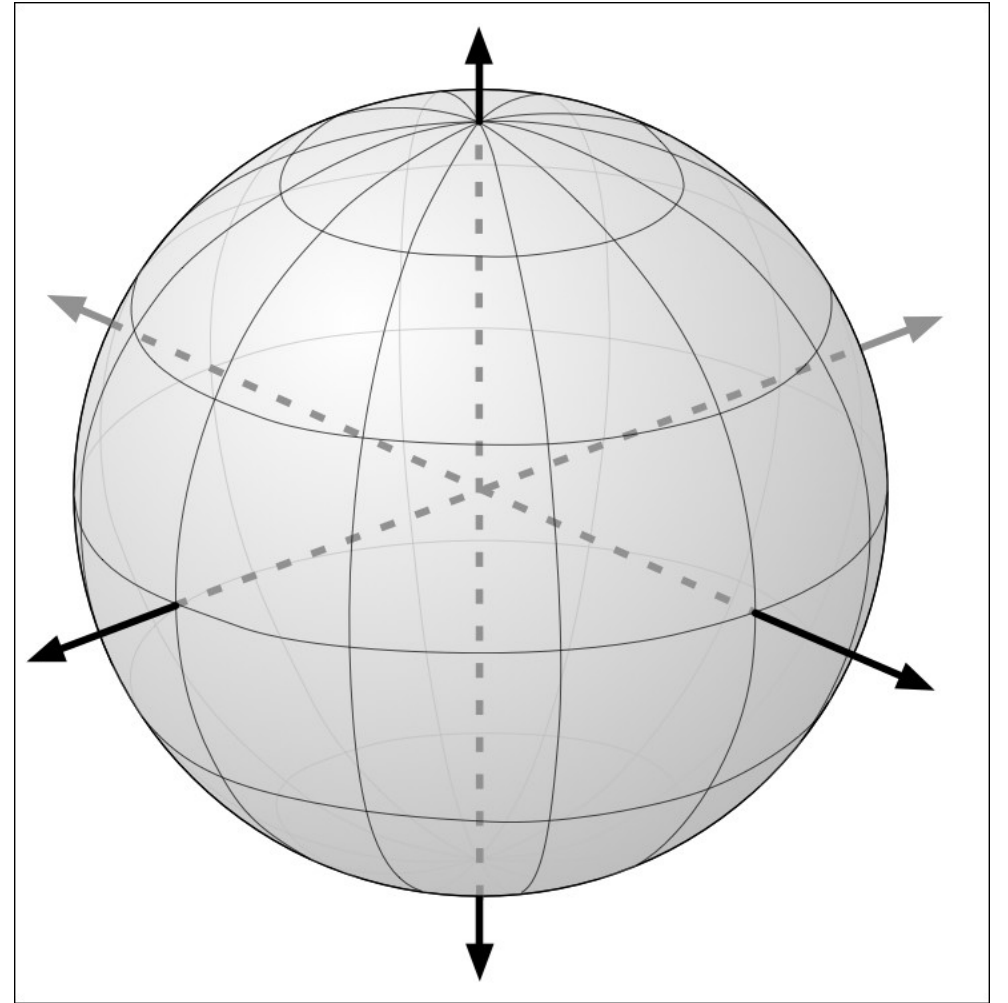
- Points on Earth's surface are measured by **latitude** (north-south) and **longitude** (east-west).

- **Coordinate System**

- The Earth is often modeled as a sphere, with **X, Y, Z axes** drawn through the center.

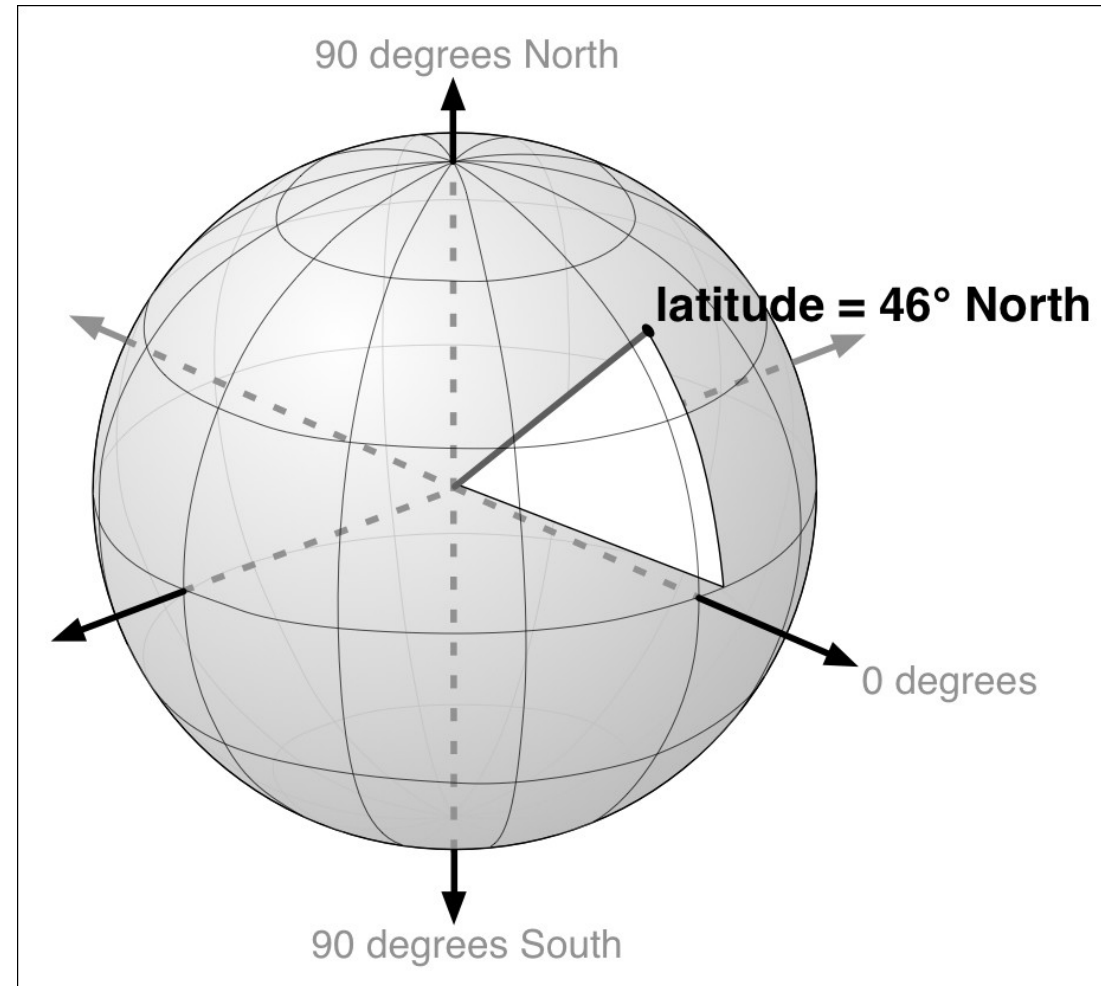
- **Latitude & Longitude Example**

- Example location: **38.167446° S, 176.234436° E**



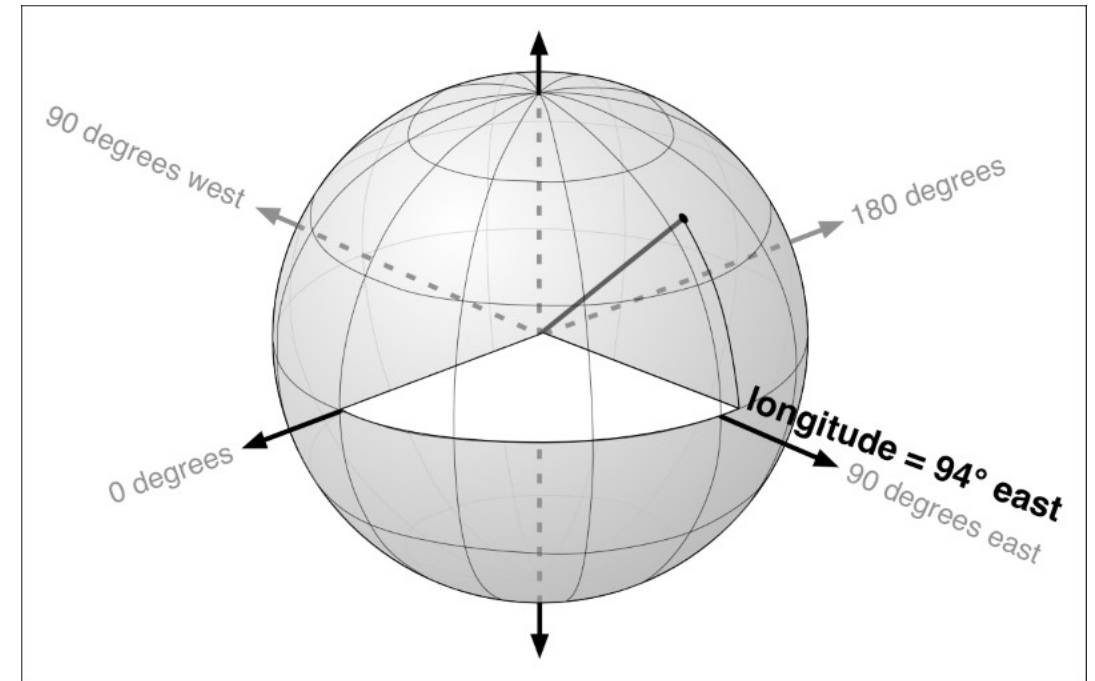
Latitude and Earth's Center

- **Connecting Point to Earth's Center:**
 - For any point on the surface, imagine a line from that point to the **center of the Earth**.
- **Latitude Angle:**
 - The **latitude** is the angle between this line and the **equator**, measured in the north-south direction.
 - Points **north** of the equator have **positive latitude**, and points **south** have **negative latitude**.



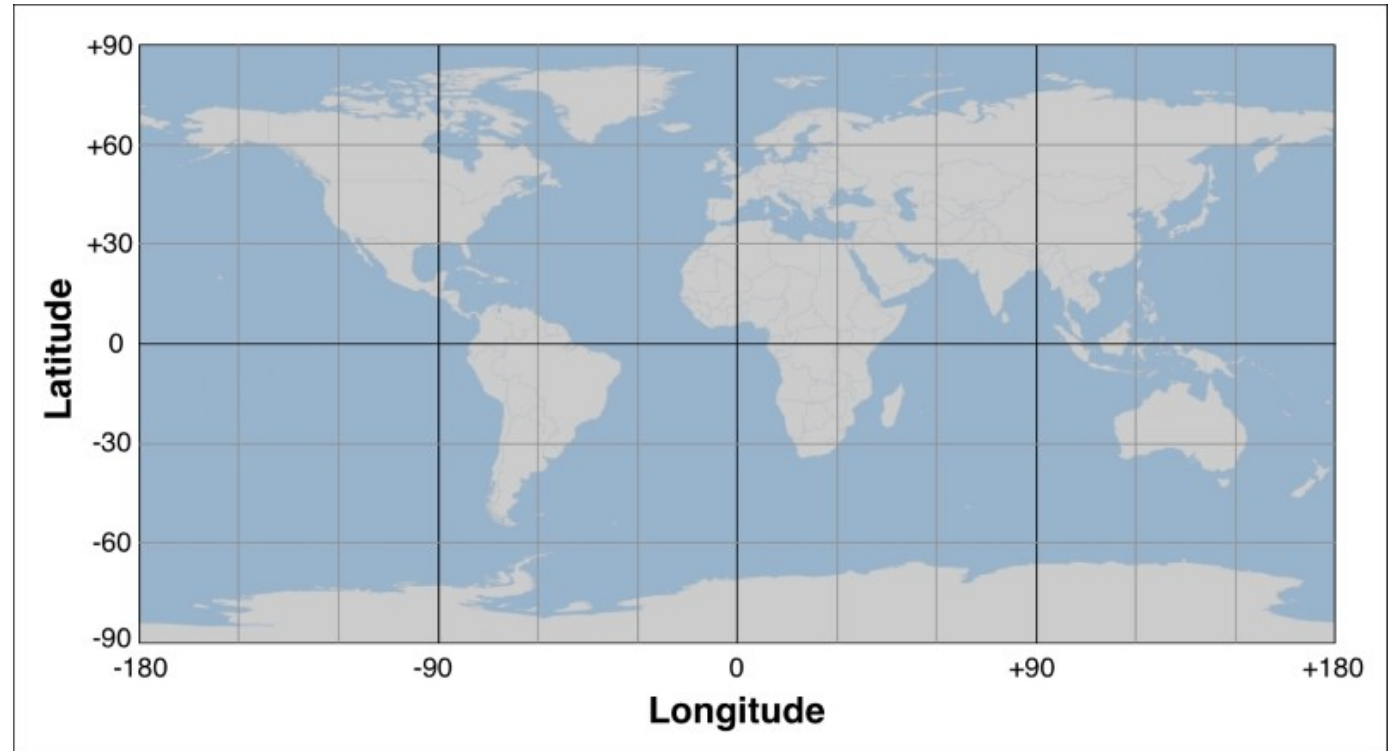
Longitude and Earth's Center

- **Longitude Angle:**
 - Similar to latitude, longitude is measured by the angle between a line from a point on the Earth's surface to its center.
- **East-West Orientation:**
 - This angle is relative to an **arbitrary starting point**, typically the **Prime Meridian** at the Royal Observatory in **Greenwich, England**.
 - Points **east** of Greenwich have **positive longitude**, and points **west** have **negative longitude**.



Latitude & Longitude Lines

- **Parallels (Horizontal Lines)**
 - Represent points of **equal latitude**.
 - The parallel at **0° latitude** is the **equator**.
- **Meridians (Vertical Lines)**
 - Represent points of **equal longitude**.
 - The meridian at **0° longitude** is the **prime meridian**.



Working with Latitude & Longitude

- **Western Longitudes:**

- Typically **negative**, but sometimes **positive** (e.g., US-specific data).

- **Longitude Wrap-Around:**

- At $\pm 180^\circ$, longitudes "wrap around" from **179° to -179°** .
- Use libraries for accurate distance calculations.

- **Types of Locations:**

- **Geodetic Location:** Precise point on Earth (latitude/longitude).
- **Civic Location:** Street address (less precise).
- **Jurisdictional Location:** Boundaries like cities or electoral wards.

Handy Notion from class -- of longitude at the equator is approximately 111 km (111,319.5m)

Latitude & Longitude Representations

- **Decimal Degrees (DD):**
 - Example: **38.167446°, -176.234436°**
 - Common in modern GPS systems and calculations.
- **Degrees, Minutes, Seconds (DMS):**
 - Example: **38° 10' 2.8056" S, 176° 14' 3.97" E**
 - More traditional, often used in navigation.

Order of Latitude & Longitude

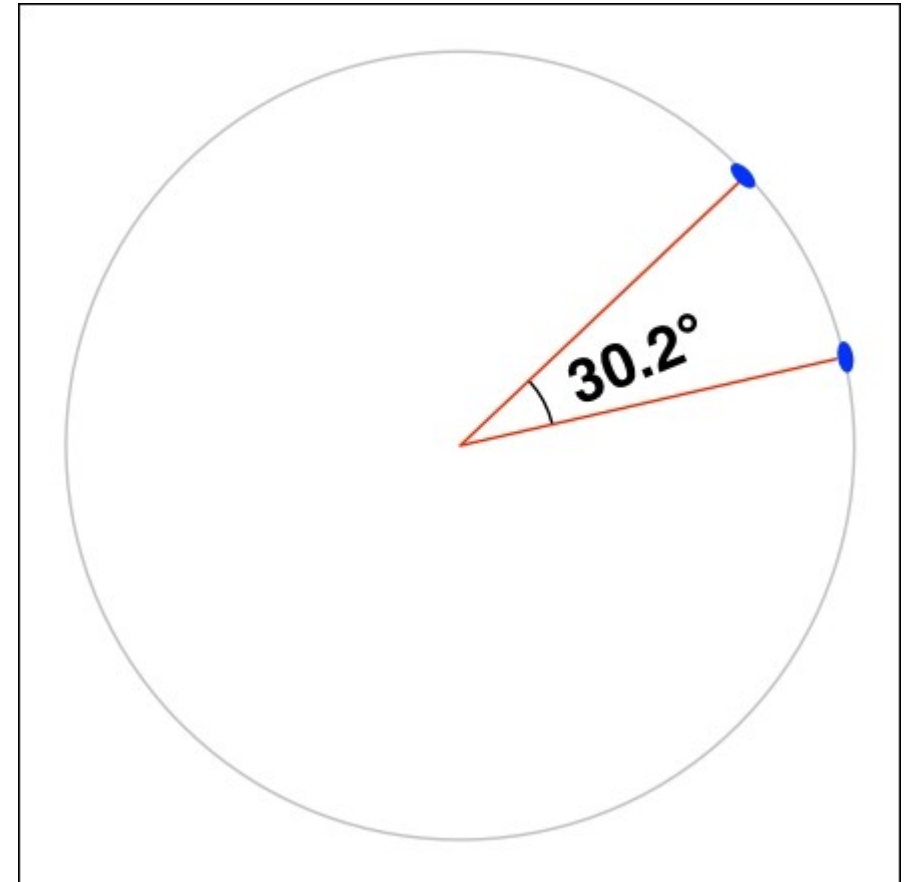
- **(Lat, Lon) Format:**
 - Common in **GIS systems** and mapping tools.
 - Example: **(38.167446°, -176.234436°)**
- **(Lon, Lat) Format:**
 - Used in some **data formats** (e.g., GeoJSON).
 - Example: ****(-176.234436°, 38.167446°)**

Latitude & Longitude: Symbols and Notation

- **Latitude (φ or Phi):**
 - North (+) or South (-) of the equator.
 - Example: 38° N (or $\varphi = 38^\circ$)
- **Longitude (λ or Lambda):**
 - East (+) or West (-) of the Prime Meridian.
 - Example: 176° W (or $\lambda = -176^\circ$)
- **Degrees, Minutes, Seconds (DMS):**
 - $^\circ$ (degrees), ' (minutes), " (seconds)
 - Example: $38^\circ 10' 2''$ N, $176^\circ 14' 4''$ W

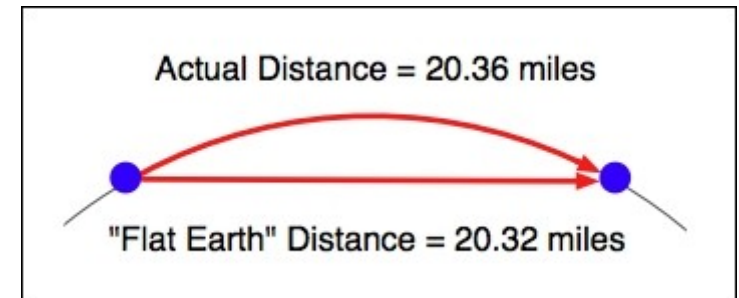
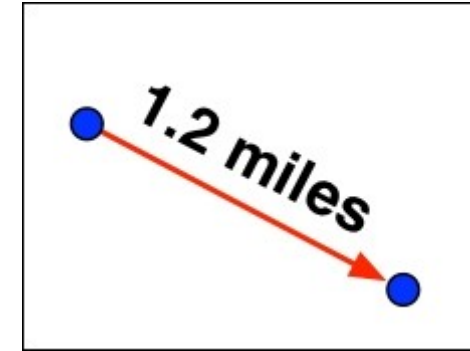
Distance on Earth's Surface

- **Types of Distance:**
 - Angular Distance
 - Linear Distance
 - Traveling Distance
- **Angular Distance**
 - **Definition:**
 - Angle between **two rays** from Earth's center through two points.
 - **Use Case:**
 - Common in **seismology** and **geospatial data**.



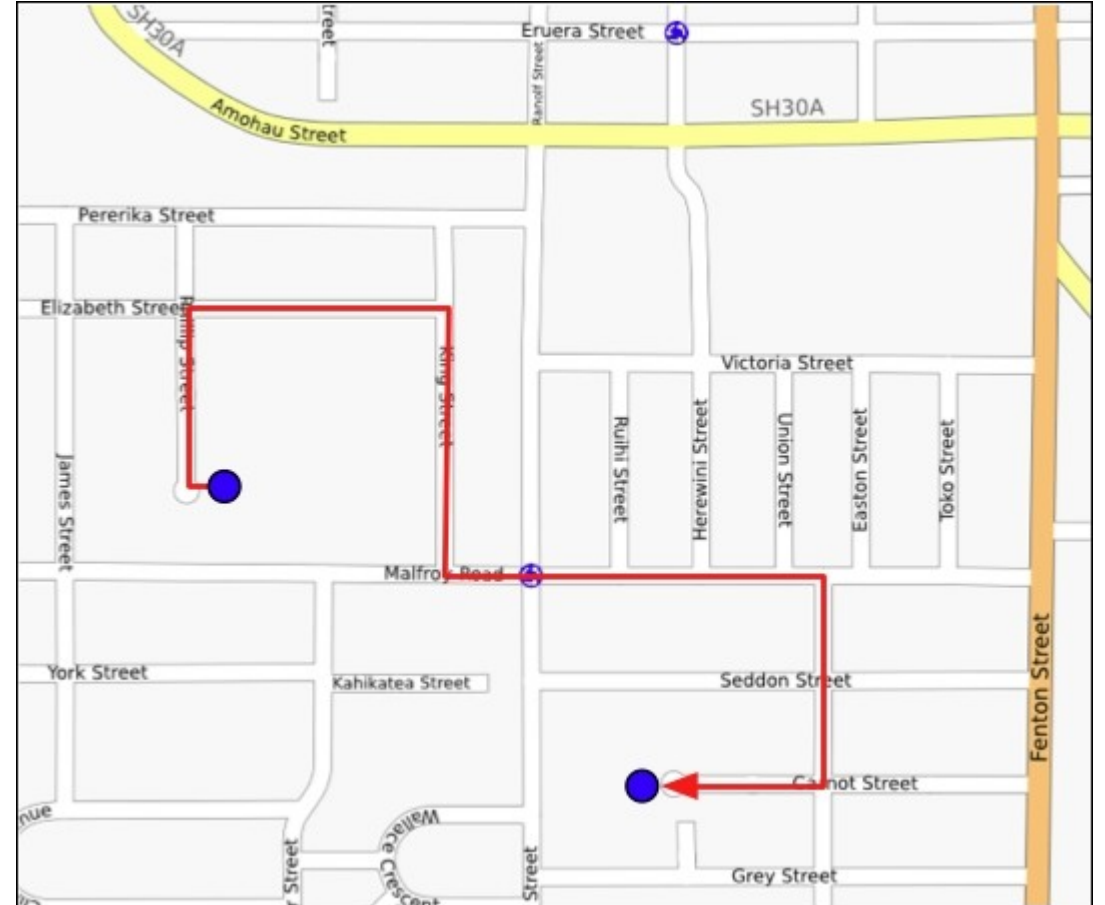
Linear Distance

- **Definition:**
 - "As the crow flies" distance between two points on Earth's surface.
- **Note:**
 - **Not as simple** due to Earth's curvature.



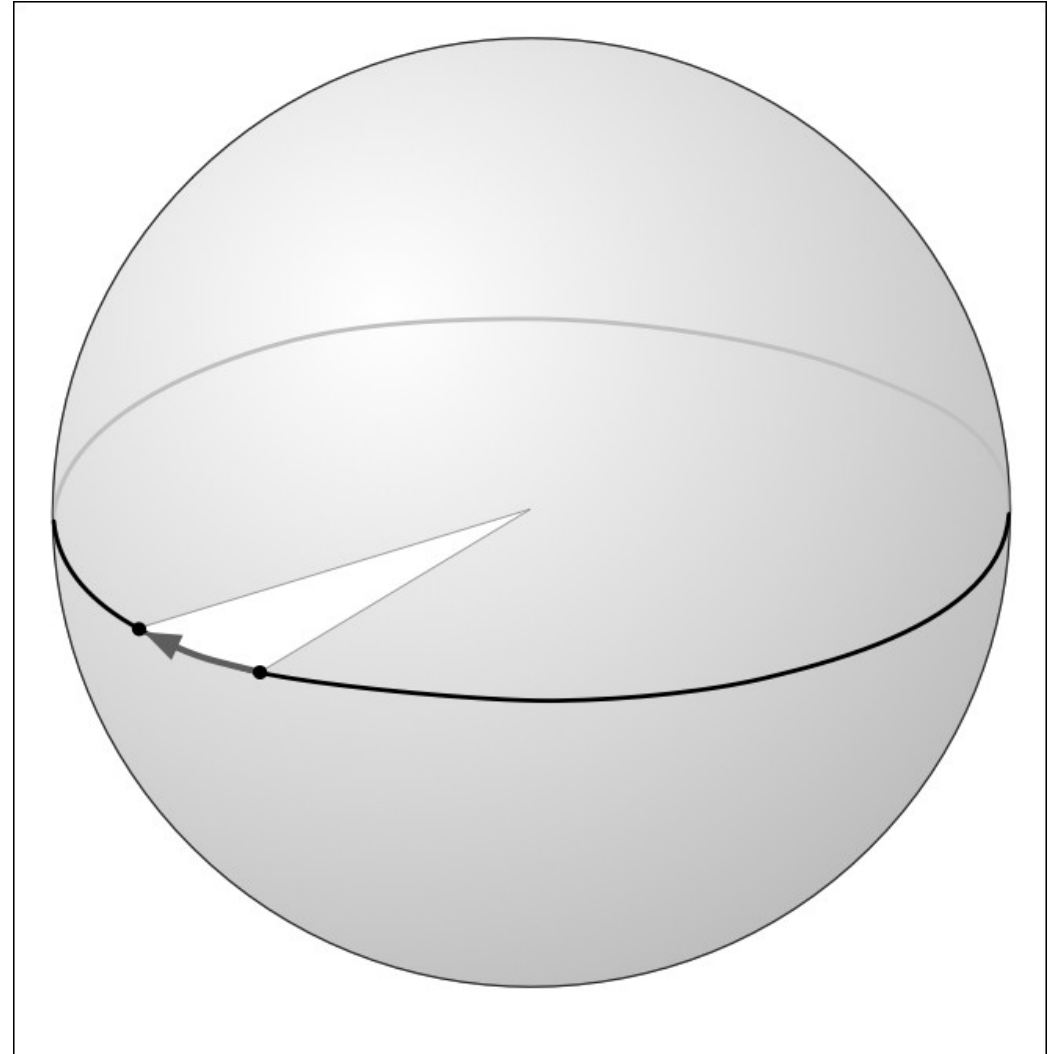
Traveling Distance

- **Definition:**
 - Actual distance traveled along roads or routes.
- **Context:**
 - Useful for **real-world navigation**, unlike straight-line distances.



Great-Circle Distance

- **Definition:**
 - Shortest distance on Earth's surface, measured along a **semicircle** centered on Earth's core.
- **Importance:**
 - Used in **geodesic calculations**.



Calculating Great-Circle Distance

- **Haversine Formula:**
 - Used to calculate great-circle distance assuming Earth is a **sphere**.
- **More Complex Techniques:**
 - Available for more **accurate** calculations considering Earth's shape.

Haversine Formula

The Haversine formula is used to calculate the great-circle distance between two points on the Earth's surface given their latitude and longitude.

This formula accounts for the Earth's curvature when computing distances.



Understanding Units in Geospatial Data

- **Critical Lesson (Mars Orbiter):**
 - Mixing **imperial** and **metric** units can cause **disasters**—like NASA's \$327M loss.
- **Distance Units:**
 - **Metric:** Millimeters, Centimeters, Meters, Kilometers
 - **Imperial:** Inches, Feet (U.S. & International), Yards, Miles (U.S. & Nautical)
- **Angular Measurements:**
 - **Degrees vs. Radians**



Remember the Mars Climate Orbiter incident from 1999?

Introduction to Map Projections

- **Projection:**
 - Converts Earth's **3D surface** into a **2D map**.
- **Distortion:**
 - **Inevitable** in projections. Each distorts **shape, area, or distance** differently.
- **Choosing a Projection:**
 - Select a projection where the **distortion** doesn't impact your **specific use case**.



Entertainment Break

<https://www.youtube.com/watch?v=7O2CBgWshiM>

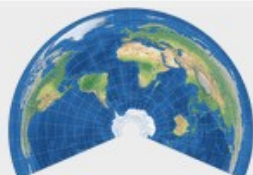


YOU LIKE ISAAC ASIMOV, XML, AND SHOES WITH TOES.
YOU THINK THE SEGWAY GOT A BAD RAP. YOU OWN 3D
GOGGLES, WHICH YOU USE TO VIEW ROTATING MODELS
OF BETTER 3D GOGGLES. YOU TYPE IN DVORAK.

Select projection to view



20/60N Equidistant Conic



20/60S Equidistant Conic



A4 Projection



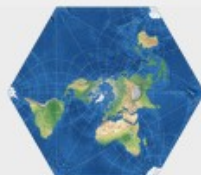
Adams Hemispheres



Adams World I



Adams World II



Adams Hexagon



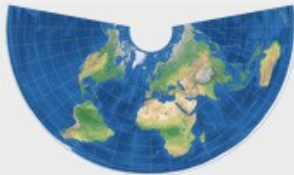
Airy (Hem.)



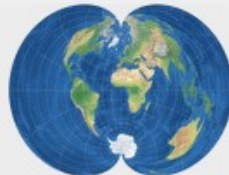
Aitoff



Aitoff-van-der-Grinten 65 [?]



Albers



American Polyconic



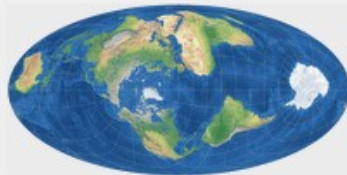
Anthracene



Apian II



Arden-Close Cylindrical



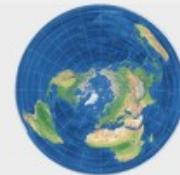
Atlantis (rotated) [?]



August Epicycloidal



Azimuthal equal-area (equat.)



Azimuthal equal-area (polar)



Azimuthal equal-area (Hem.)



Azimuthal Equidistant (equat.)



Azimuthal Equidistant (polar)



Azimuthal Equidistant (Hem.)



Bacon

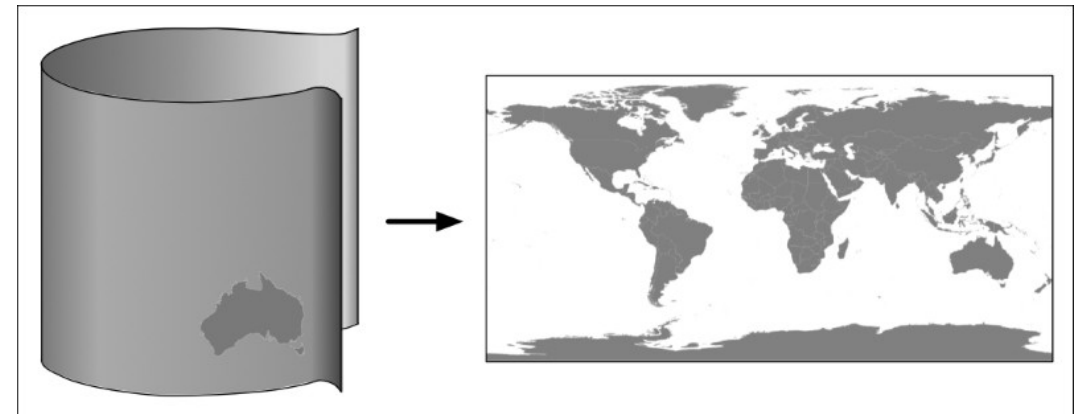
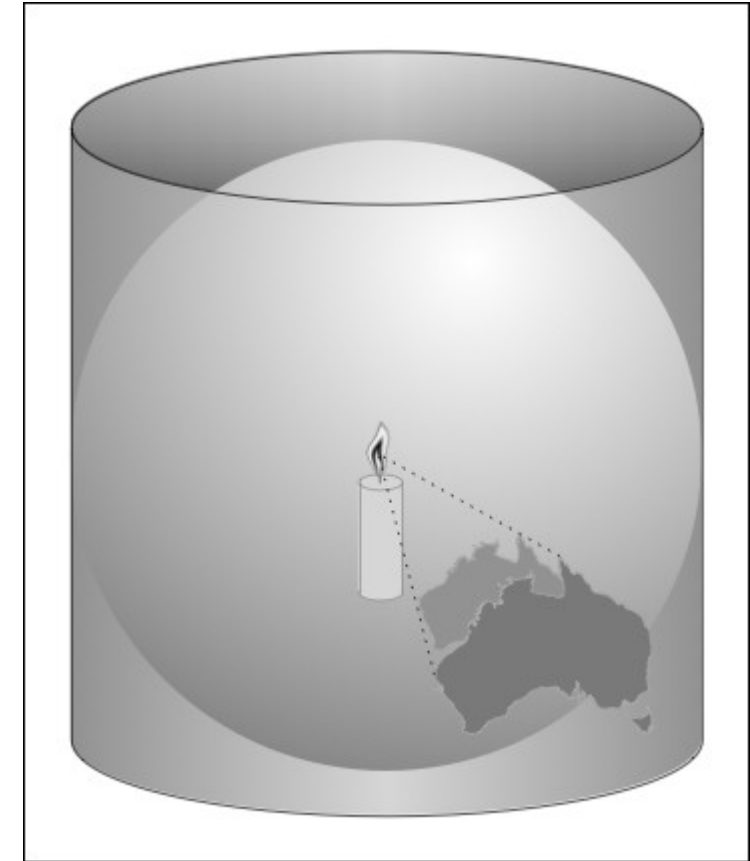


Baker Dinomic



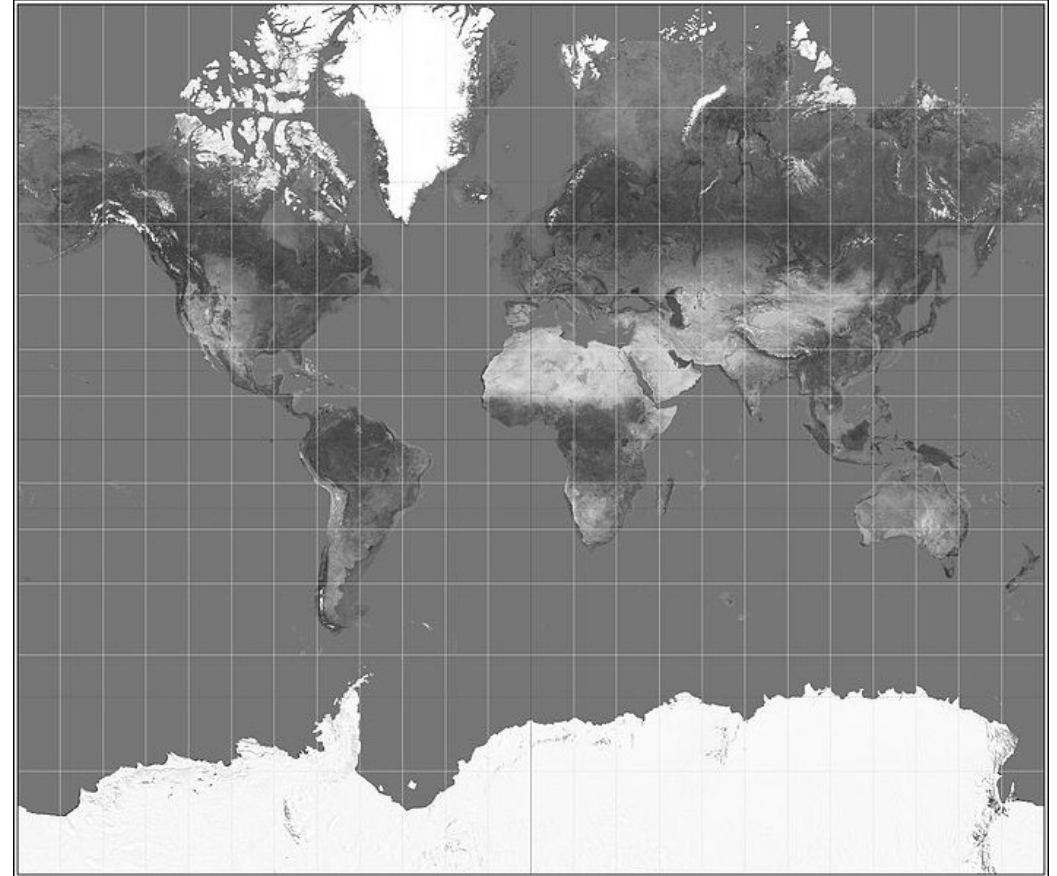
Cylindrical Projections

- **Visualizing Cylindrical Projections:**
 - Imagine Earth as a **spherical lantern** with a **candle inside**.
 - Place the lantern inside a **paper cylinder**.
 - The Earth's surface is **projected** onto the **cylinder**.
- **Unwrapping the Cylinder:**
 - **Unwrap** the cylinder to create a **2D map**.
- **Reality:**
 - In practice, **mathematical transformations** are used instead of light projection.



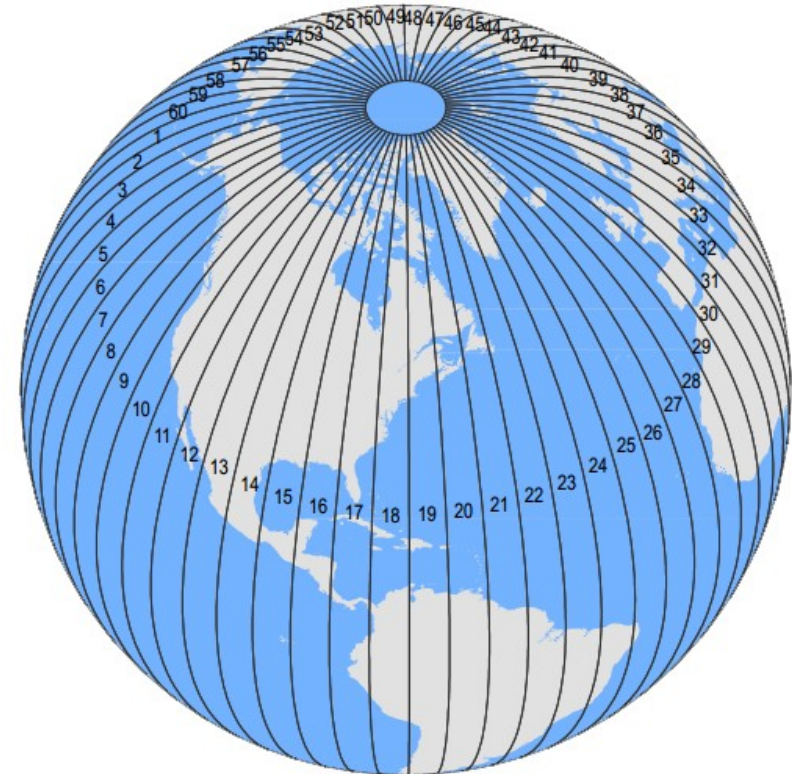
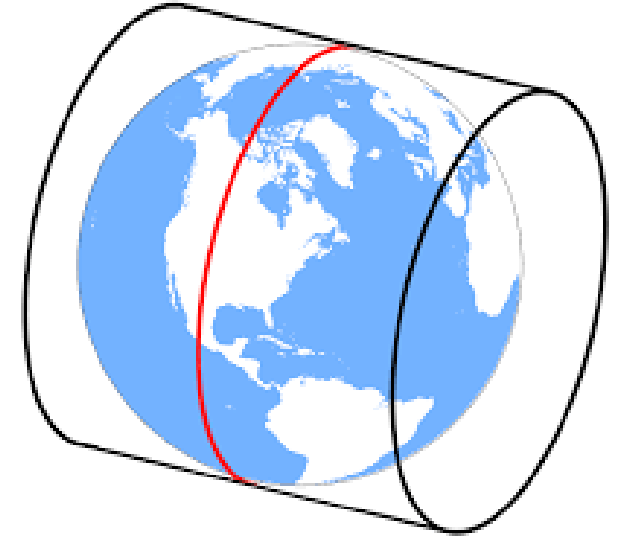
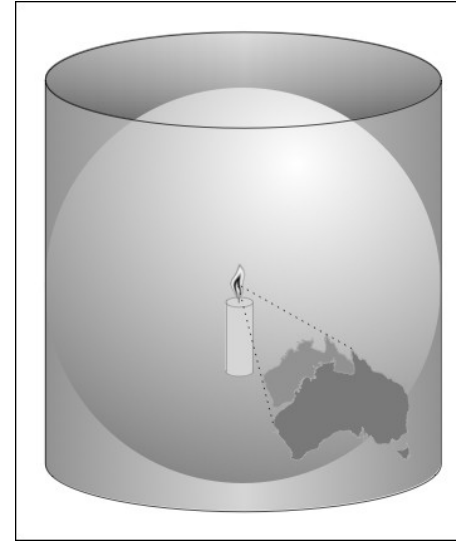
Types of Cylindrical Projections

- **Common Cylindrical Projections:**
 - **Mercator Projection:**
 - Preserves **shape**, distorts **area** (e.g., Greenland appears larger).
 - **Equal-Area Cylindrical Projection:**
 - Preserves **area**, distorts **shape**.
 - **Universal Transverse Mercator (UTM):**
 - Divides Earth into **zones** for accurate local mapping.
- **Example:**
 - **Mercator Projection** (image from Wikipedia)



UTM Projections

- Mercator Projections are least-distorted at the equator.
- A *Transposed* Mercator uses a sideways cylinder. Less projection along a line of longitude.
- UTM divides earth into 6 degree zones (60 of them)
- **Very** commonly used



Cylindrical Examples



Arden-Close Cylindrical



Balthasart



Behrmann



Braun Perspective



Braun Stereographic



BSAM Cylindrical



Cassini (rotated)



Central Cylindrical



Gall Isographic



Gall-Peters



Gall Stereographic



Hobo-Dyer



Kavraiskiy I



Kharchenko-Shabanova



Lambert Cylindrical



Mercator



Miller



Miller, Compact



Miller Perspective Compromise



Patterson



Equirectangular (0°)



Equirectangular (28°)



Equirectangular (35.6°)



Equirectangular (60°)



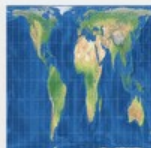
Smyth equal-surface



Tobler Cylindrical 1



Tobler Cylindrical 2



Tobler's World in a Square



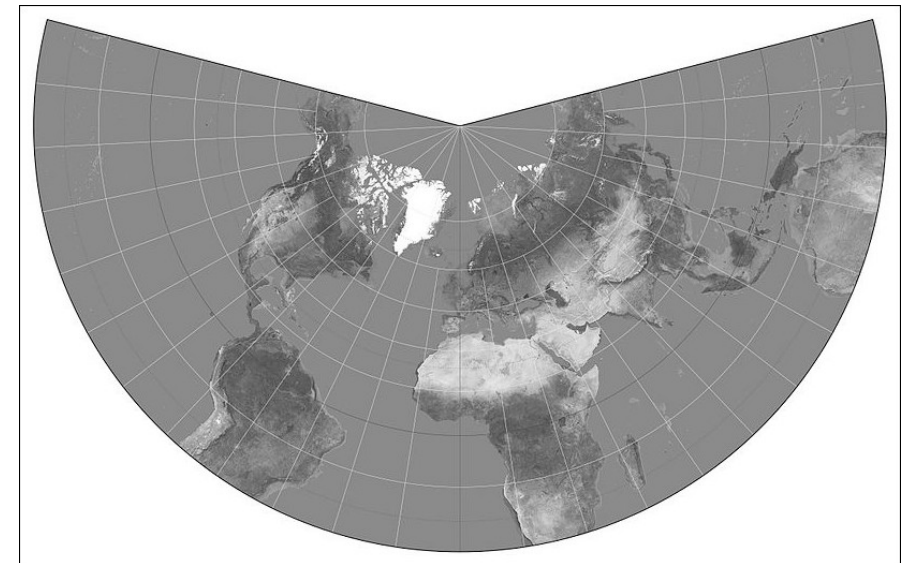
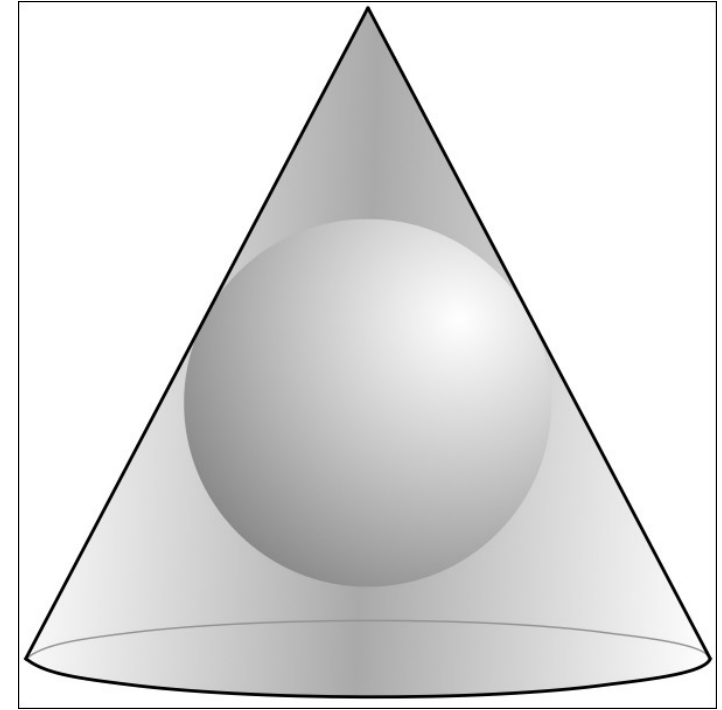
Trystan Edwards



Urmayev II Cylindrical

Conic Projections

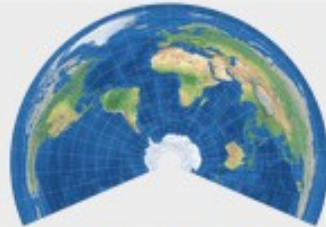
- **Conic Projection:**
 - Earth's surface is projected onto a **cone**.
- **Unwrapping the Cone:**
 - The cone is **unwrapped** to form a **2D map**.
- **Common Types:**
 - **Albers Equal-Area Projection:**
 - Preserves **area**.
 - **Lambert Conformal Conic Projection:**
 - Preserves **shape**, distorts **size**.
 - **Equidistant Projection:**
 - Preserves **distance** along specific lines.
- **Use Case:**
 - **Polar-aligned conic projections** are ideal for areas that are **wide** but **not tall** (e.g., **Russia**).



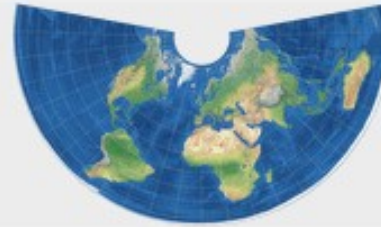
Conic Projections (Examples)



20/60N Equidistant Conic



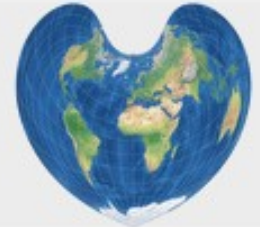
20/60S Equidistant Conic



Albers



American Polyconic



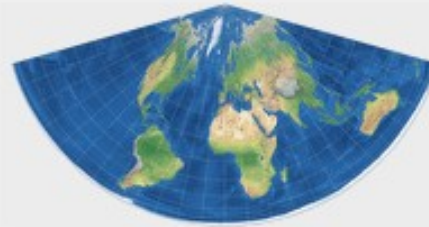
Bonne



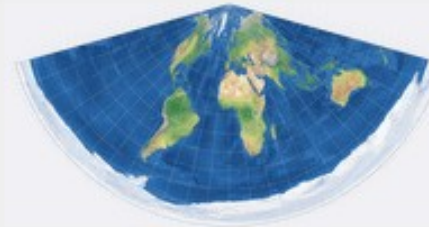
Bottomley



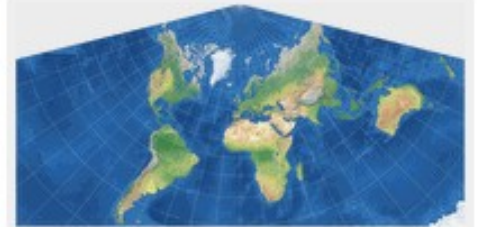
Bottomley 50°N



CM Equal-Area Conic



CM Equidistant Conic



Lambert CC



Lambert Equal-Area Conic



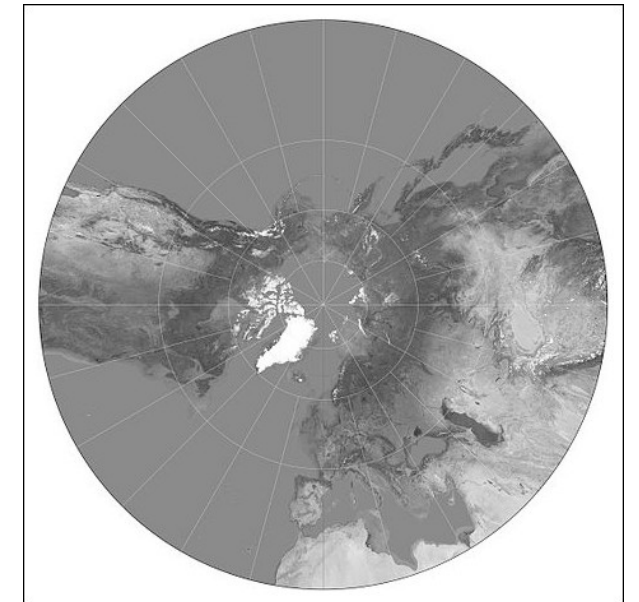
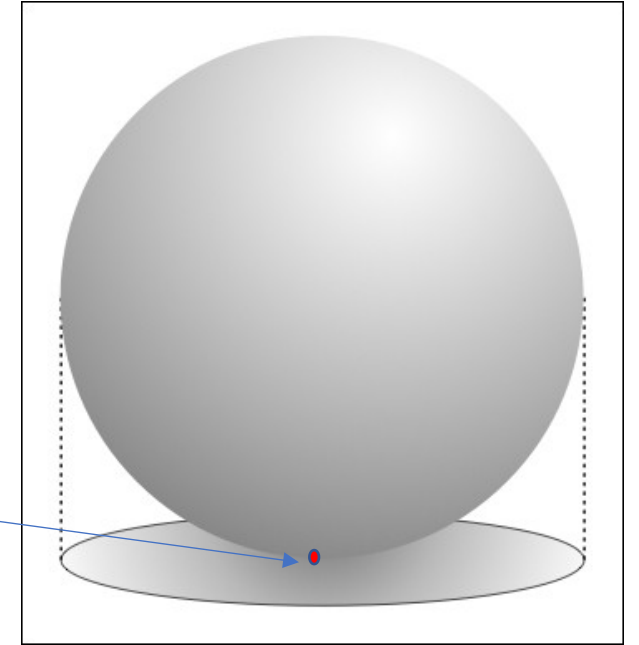
Schjerner I



Stab-Werner

Azimuthal Projections

- **Less Distortion:**
 - Minimal distortion near the **central point**; increases with distance.
- **Why "Azimuthal":**
 - Refers to the **angle** from the central point.
- **Types:**
 - **Gnomonic:** Great circles appear as straight lines.
 - **Lambert Equal-Area:** Preserves area.



Azimuthal Examples



Airy (Hem.)



Azimuthal equal-area (equat.)



Azimuthal equal-area (polar)



Azimuthal equal-area (Hem.)



Azimuthal Equidistant (equat.)



Azimuthal Equidistant (polar)



Azimuthal Equidistant (Hem.)



Breusing Geometric



Breusing Harmonic



Ginzburg I



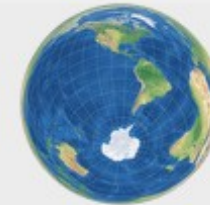
Ginzburg II



Gott-Mugnolo Azimuthal (equat.)



Gott-Mugnolo Azimuthal (Hem.)



Spilhaus-Lambert equal-area azi.

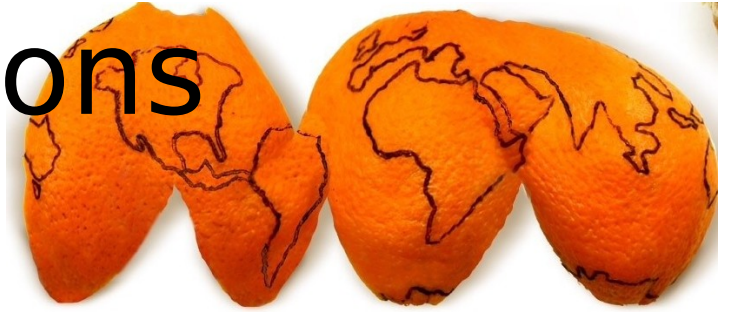


Spilhaus Stereographic



Stereographic (Hem.)

The Nature of Map Projections



- **No Perfect Projection:**
 - Every projection **distorts** Earth's surface.
- **Gauss' Theorem:**
 - Proved that projecting a **sphere** onto a flat plane always causes distortion.
- **Purpose-Driven Projections:**
 - Different projections serve different **purposes**; none can do everything.
- **Why It Matters:**
 - Knowing the **projection** is critical for accurate **plotting** and **calculations** in geospatial data.

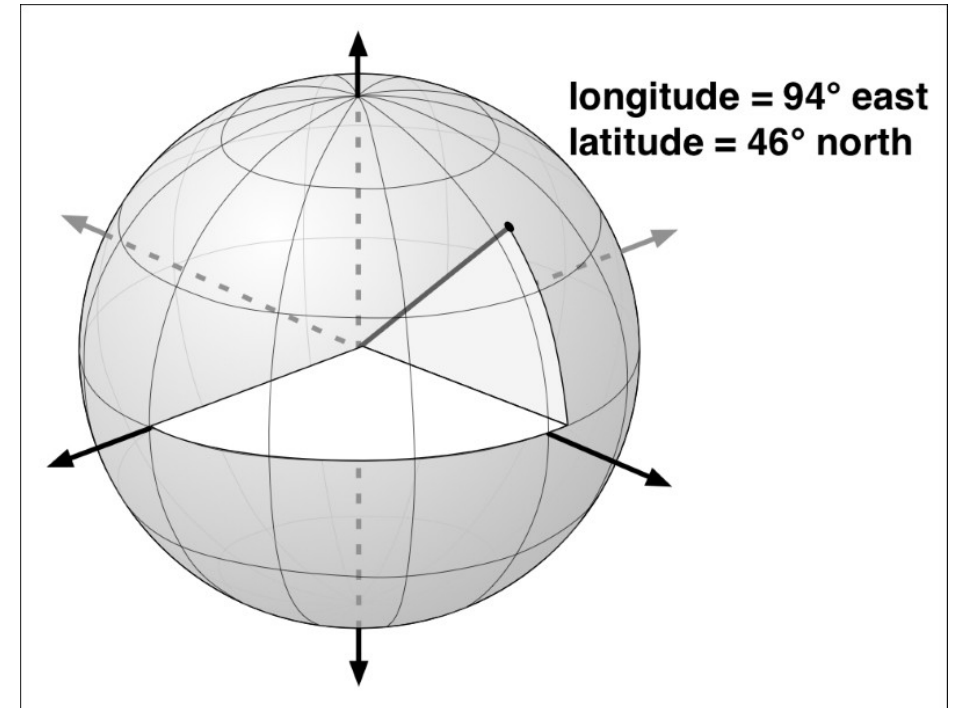


Introduction to Coordinate Systems

- **Types of Coordinate Systems:**
 - Unprojected (e.g., Latitude/Longitude)
 - Projected (e.g., UTM)
 - 3D Cartesian (e.g., ECEF, NED)
- **Purpose:**
 - Coordinate systems define **locations** on Earth's surface in either **3D space** or **2D projections**.

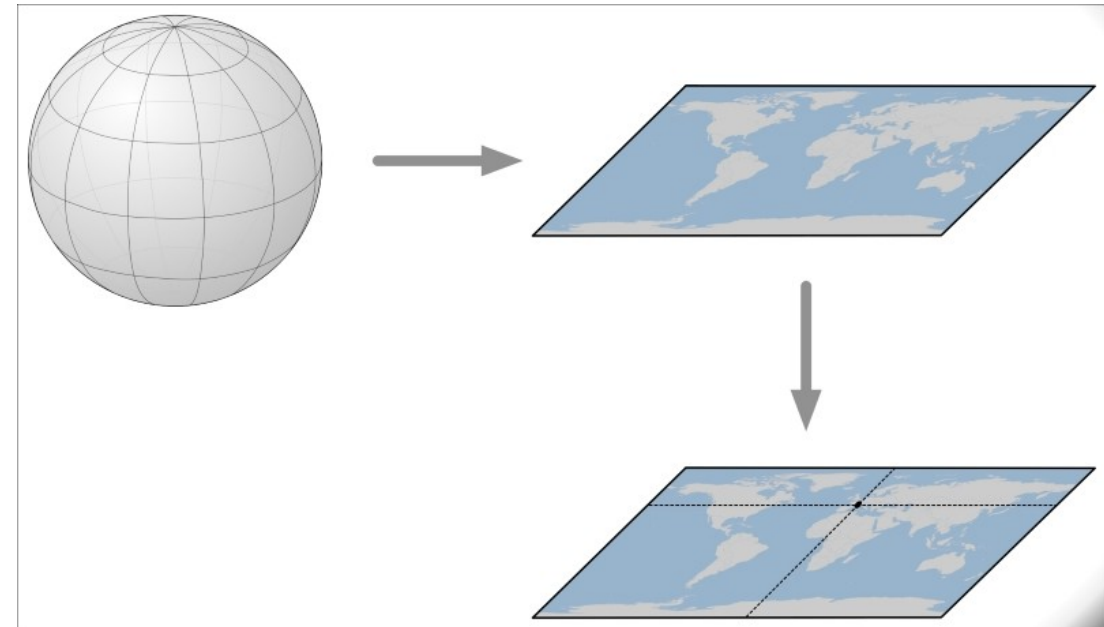
Unprojected Coordinate Systems

- **Example:**
 - Latitude/Longitude
- **Characteristics:**
 - Directly represents **points on Earth's surface**.
 - Useful for **global coverage**.
- **Challenge:**
 - **Difficult** for geospatial calculations (e.g., distance, area).



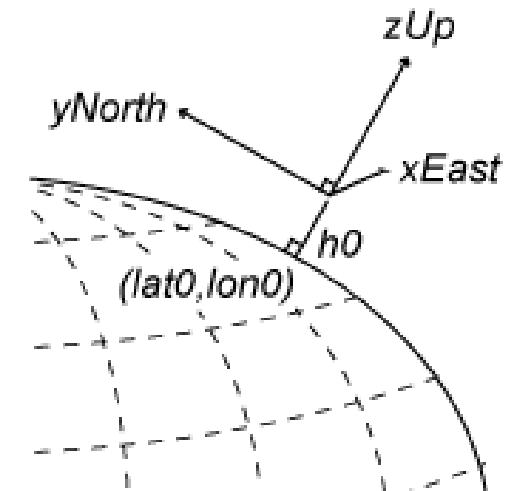
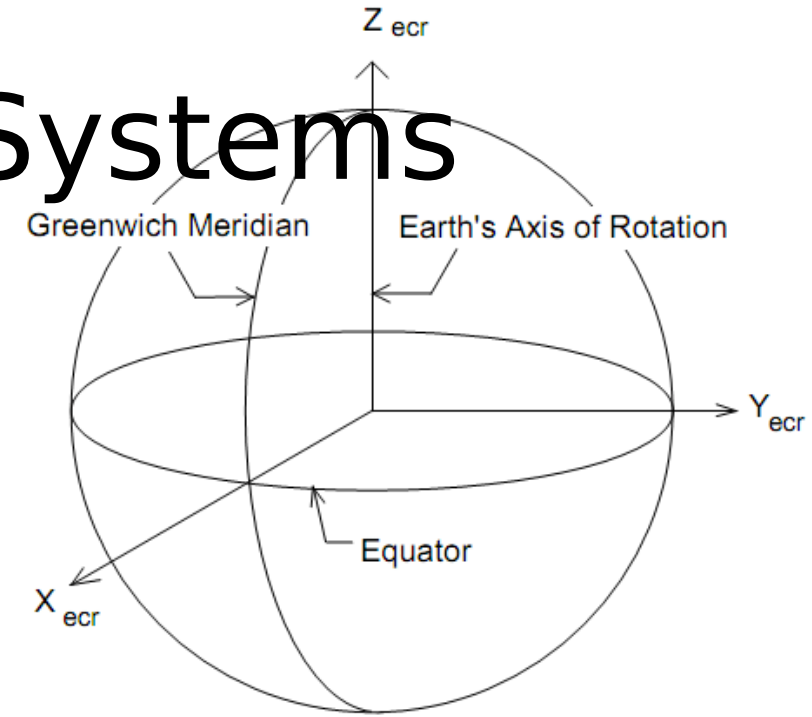
Projected Coordinate Systems

- **Example:**
 - UTM (Universal Transverse Mercator)
 - **Characteristics:**
 - Converts Earth's surface into a **2D Cartesian plane**.
 - Easier for **distance** and **area calculations**.
 - **Key Requirement:**
 - Must know the **map projection** used to create the coordinate system.
- (Also conversion always has some distortion!)



3D Cartesian Coordinate Systems

- **Examples:**
 - ECEF (Earth-Centered, Earth-Fixed)
 - NED (North-East-Down)
 - ENU (East-North-Up)
- **Characteristics:**
 - Represents points in a **3D space** relative to Earth's center.
 - Useful for **global positioning** and **navigation**.
- **Benefit:**
 - Allows for **precise calculations** in 3D space.
 - Cartesian coordinates with no distortion



Coordinate System Reference Points

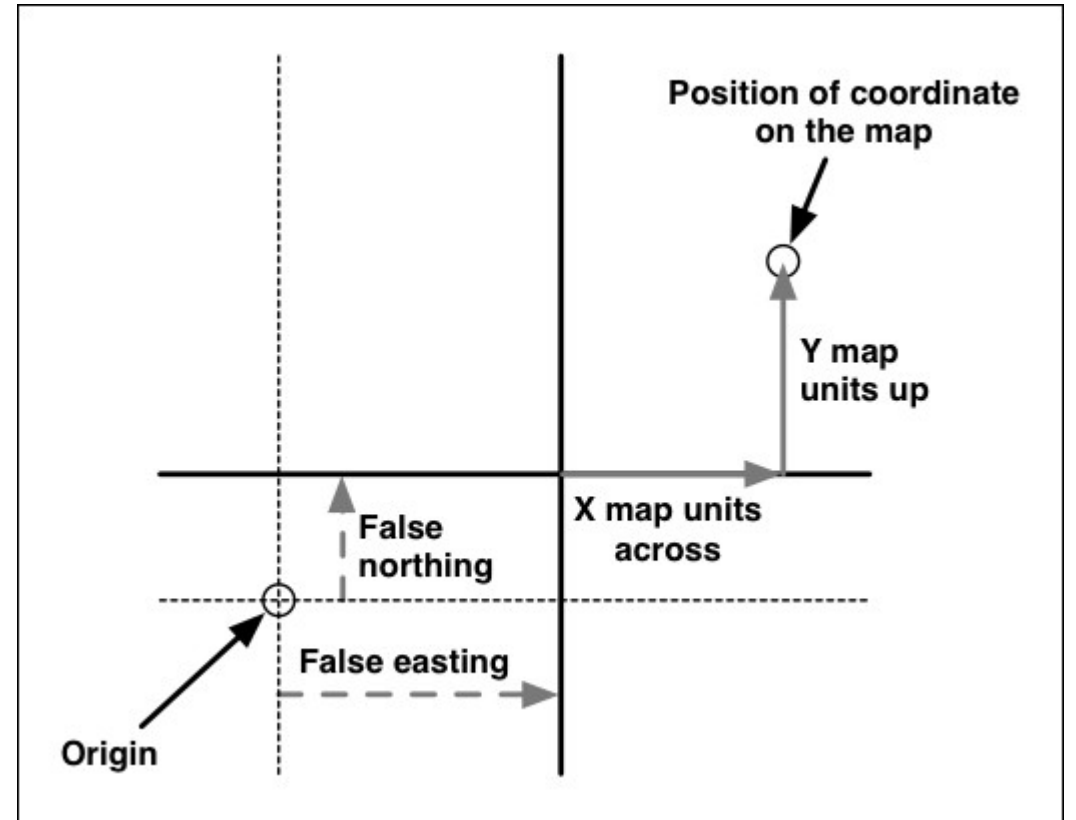
- **Unprojected Systems:**

- **Lat/Long:**

- **Longitude 0°:** Prime Meridian (Greenwich, England)
 - **Latitude 0°:** Equator

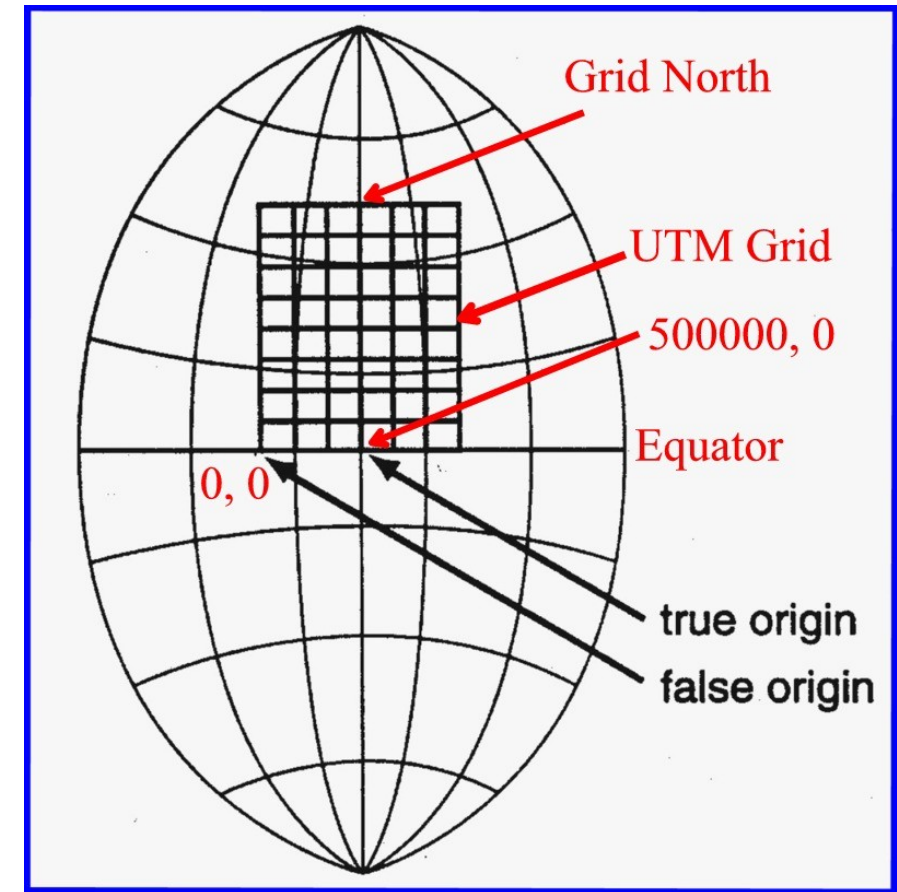
- **Projected Systems:**

- Defined by **origin** and **map units**.
 - May use **false northing** and **false easting** to adjust the origin.
 - (Mainly to avoid negative values)



Universal Transverse Mercator (UTM) (again)

- **Divides Earth into 60 Zones:**
 - Each zone covers **6 degrees of longitude**.
- **Zone-Specific Projections:**
 - Each zone uses a unique **map projection** to minimize distortion.
- **Coordinates Measured in Meters:**
 - **False easting/northing** ensures **positive values** within the zone.
- **Caution:**
 - Avoid calculating distances across **different zones** due to distortion.



Advantages of Projected Coordinates

- **Simple Calculations:**
 - **Distance** and **area** easily computed using **Cartesian coordinates**.
- **Example:**
 - In the **UTM system**, distances between two points are measured in **meters**, simplifying geospatial analysis.
- **Limitations:**
 - Only accurate over a **small area** (e.g., within the same UTM zone).

When to Use Unprojected Coordinates

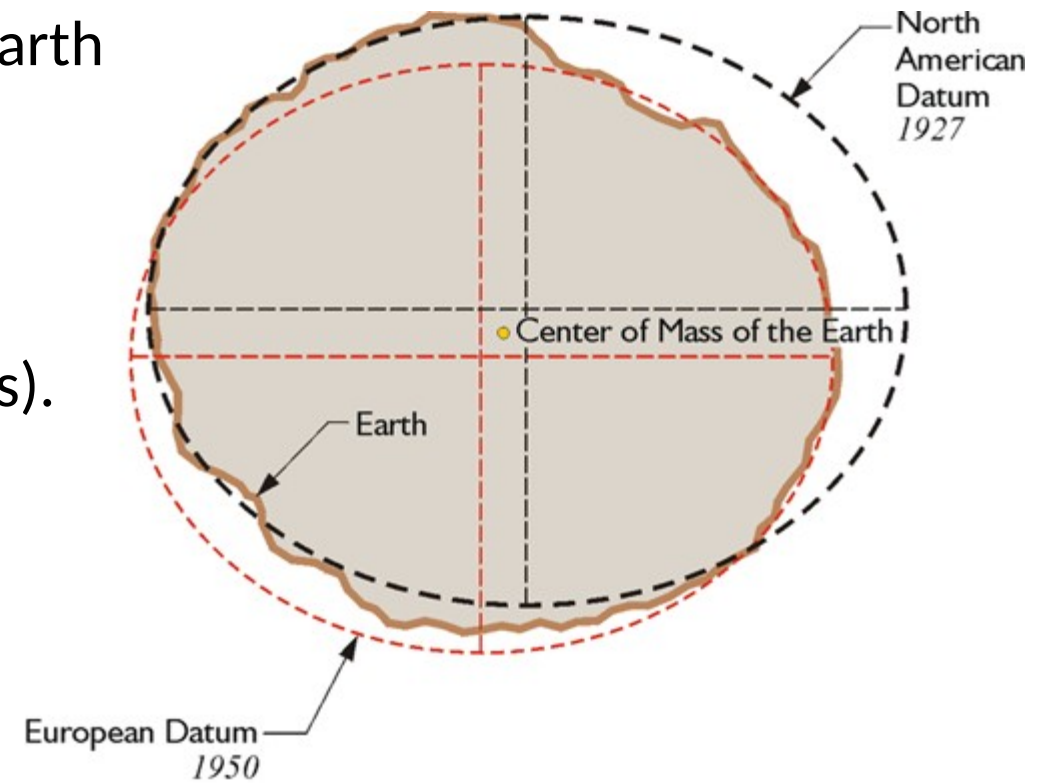
- **Global Coverage:**
 - Useful for **representing locations** anywhere on Earth.
- **Flexible Across Zones:**
 - No need to worry about **crossing UTM zones**.
- **Challenge:**
 - More complex for **distance and area calculations**.

Key Takeaways on Coordinate Systems

- **Know the System:**
 - Always be aware of the **coordinate system** in use (projected or unprojected).
- **Match Projections:**
 - Ensure all data uses the **same projection** for accurate calculations.
- **Context Matters:**
 - Use **projected coordinates** for local calculations and **unprojected coordinates** for global coverage.
- **Conversions:**
 - Be prepared to **convert** between systems when necessary.

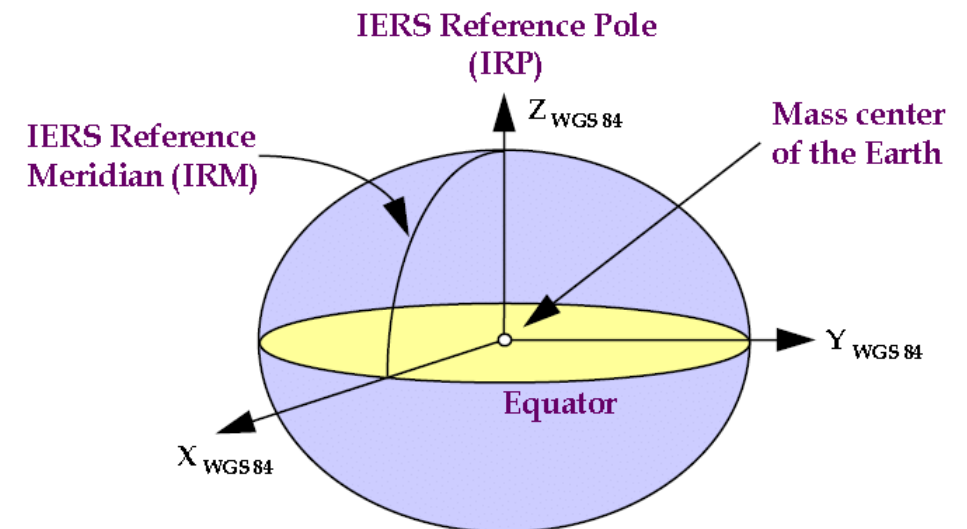
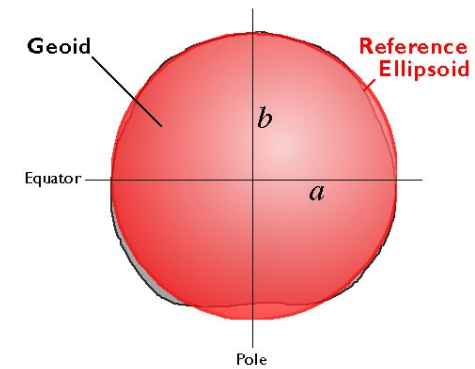
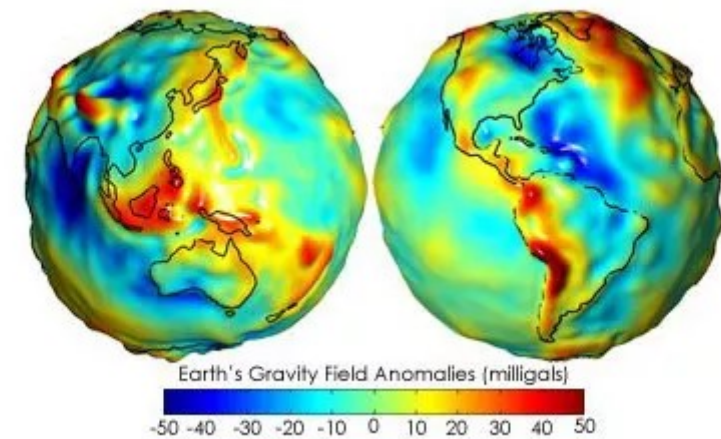
Introduction to Datums

- **Definition:**
 - A **datum** is a **mathematical model** of the Earth used to describe locations.
- **Components:**
 - Set of **reference points**.
 - Model of Earth's **shape** (used in projections).
- **Purpose:**
 - Essential for both **map projections** and **coordinate systems**.



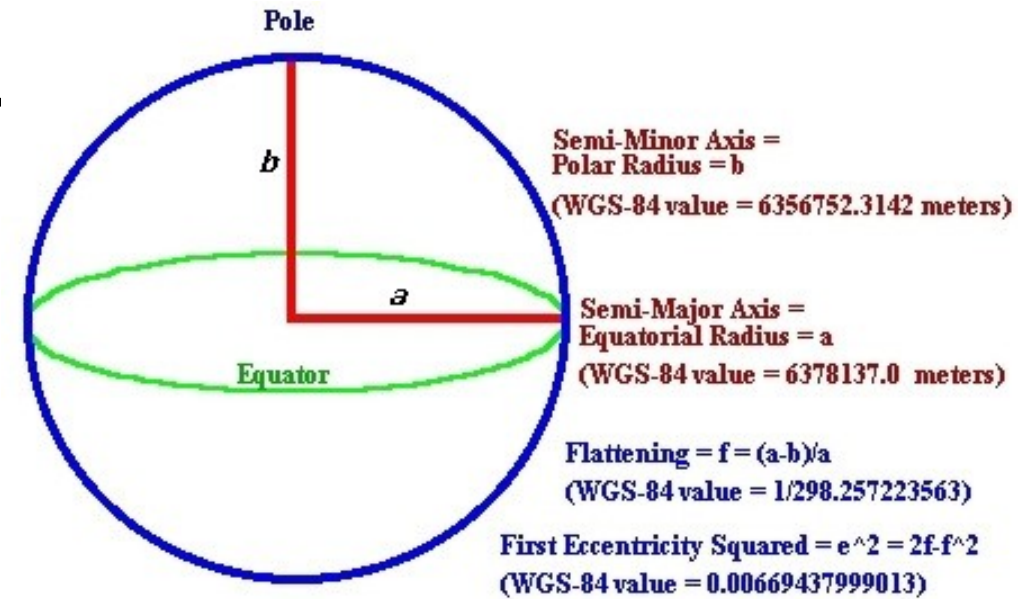
Common Datums

- **NAD 27 (North American Datum 1927):**
 - Uses **Clarke Spheroid 1866**.
 - Reference point: **Meades Ranch, Kansas**.
 - Local to **North America**.
- **NAD 83 (North American Datum 1983):**
 - Uses **GRS 80** model.
 - Covers **U.S., Canada, Mexico, Central America**.
- **WGS 84 (World Geodetic System 1984):**
 - Uses **EGM 96** model.
 - **Global datum** used by **GPS** satellites.
 - **Similar to NAD 83** in the U.S.



WGS 84 Ellipsoid Form

- **Ellipsoid Parameters:**
- **Semi-major axis (a):**
 - Equatorial radius: 6,378.137 km
- **Semi-minor axis (b):**
 - Polar radius: 6,356.752 km
- **Flattening Factor (f):**
- **Ellipsoid Formula:**
 - The equation for an ellipsoid:
- **Purpose:**
 - Models Earth's **shape** for precise global positioning.

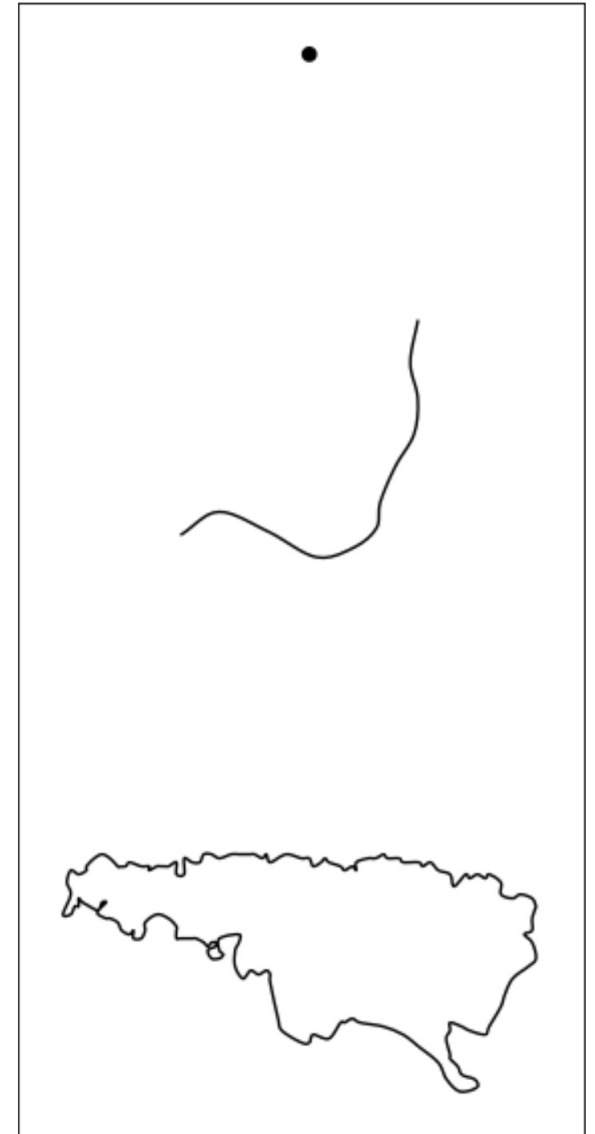


Why Datums Matter

- **Impact on Location Accuracy:**
 - A point in **NAD 27** may be **hundreds of feet** away from the same point in **WGS 84**.
- **Critical for Data Integrity:**
 - Always **know the datum** used for geospatial data.
- **Conversion Required:**
 - Convert coordinates between **different datums** when necessary to ensure **accuracy**.

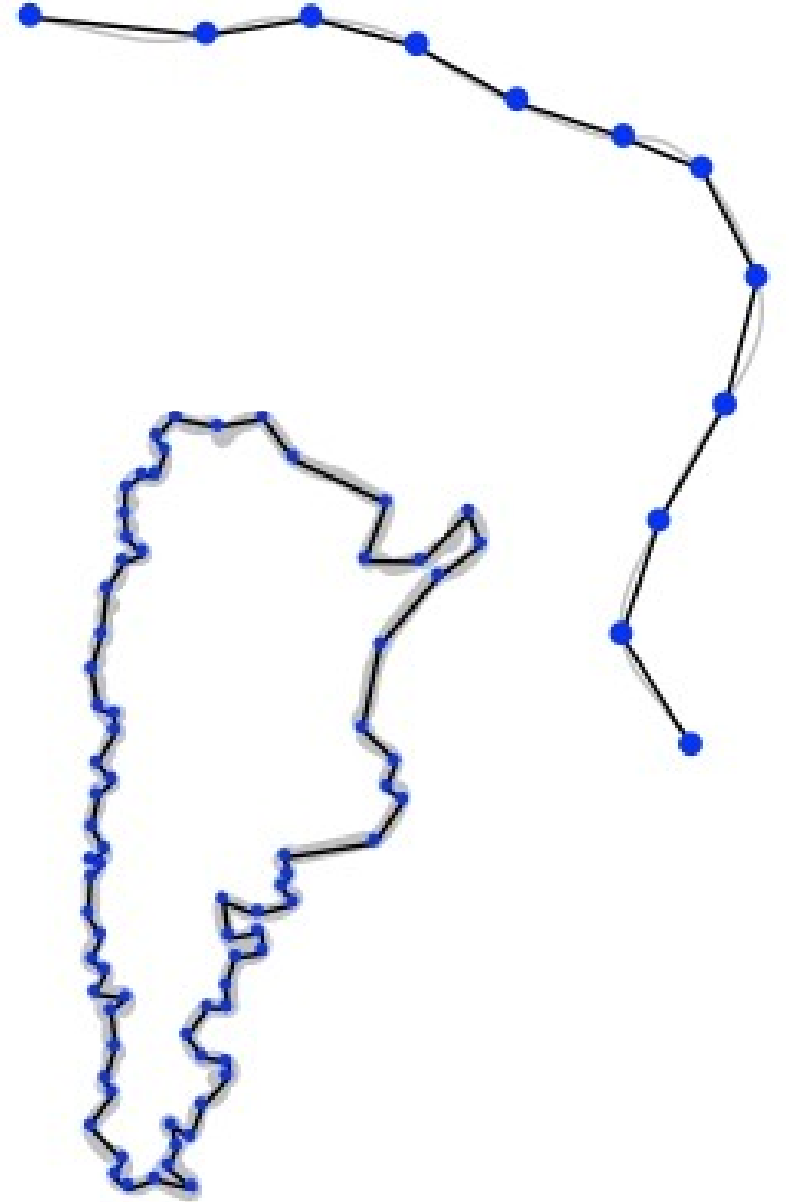
Geospatial Shapes

- **Points:**
 - Single **coordinate** in a system (e.g., lat/long).
- **Paths (LineStrings):**
 - **Connected line segments** representing roads, rivers, or boundaries.
 - Approximate curves with simple line segments.
- **Outlines (Polygons):**
 - **Closed shapes** representing areas like countries or lakes.
 - Include **exterior rings** (borders) and optional **interior rings** (holes).



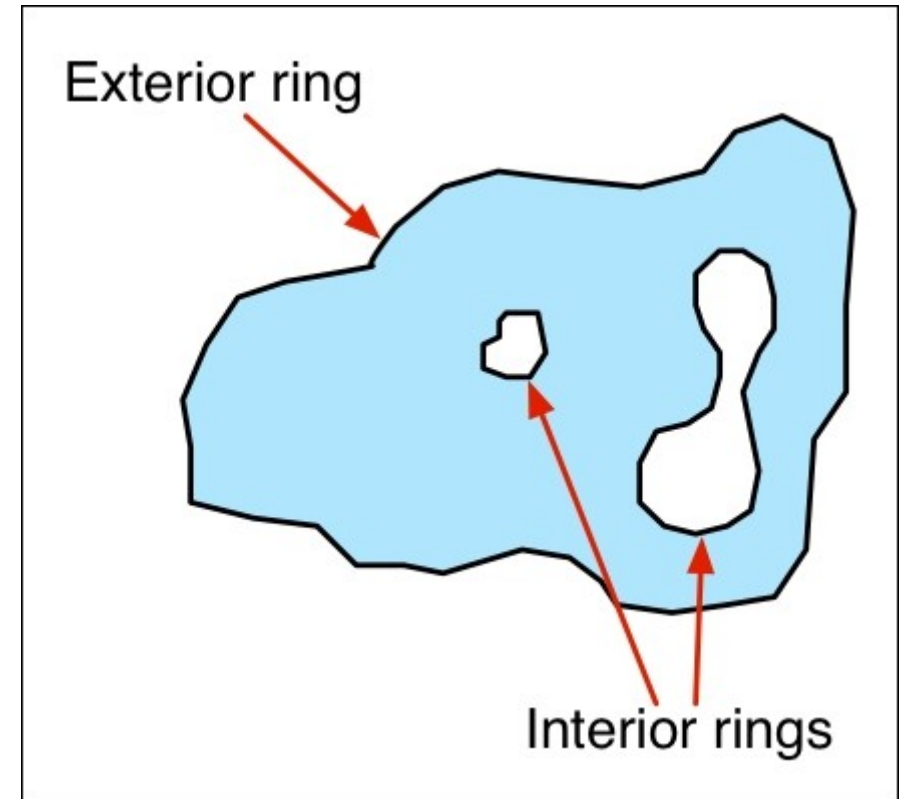
LineStrings and Polygon

- **LineStrings:**
 - Series of connected line segments forming paths.
 - Used for roads, rivers, contour lines.
- **Polygons:**
 - Closed LineStrings defining areas like countries, lakes.
 - Can contain **holes** (interior rings) for features like islands.
- **Key Term:**
 - **Linear Ring:** A closed LineString often used in polygons.



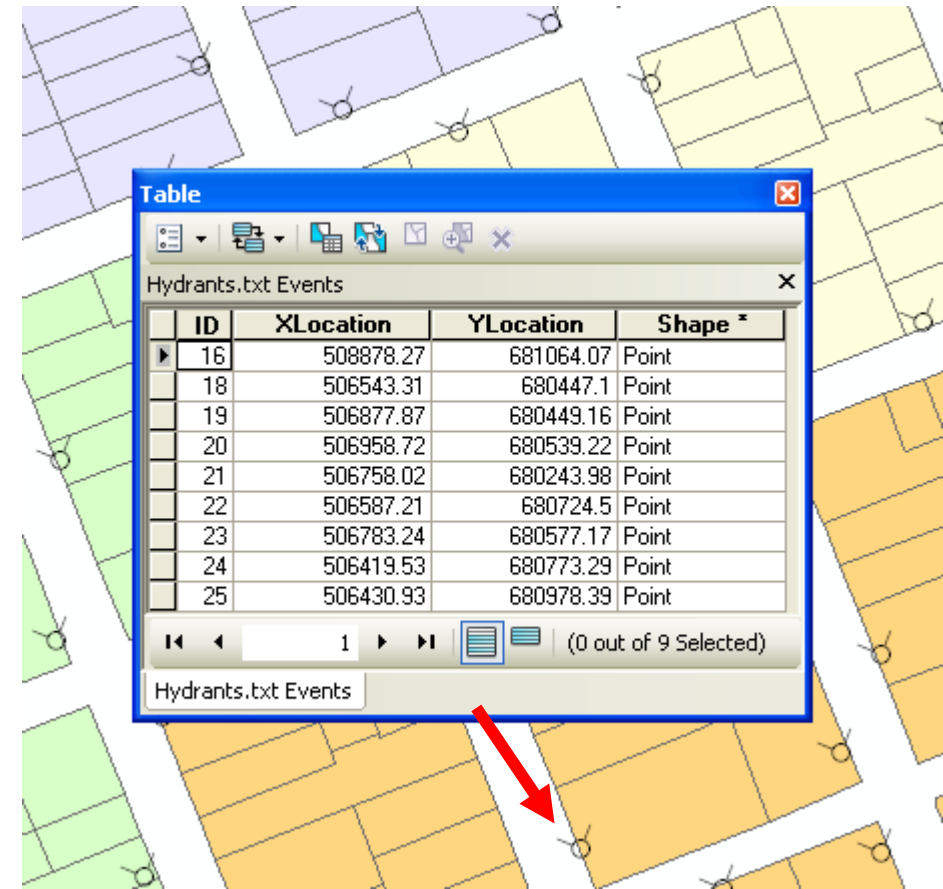
Understanding Exterior and Interior Rings

- **Exterior Ring:**
 - The **outer boundary** of a polygon, defined by a **closed LineString**.
 - **End point** = **Starting point** to form a **closed loop**.
- **Interior Rings:**
 - Represent **"holes"** within the polygon (e.g., islands within a lake).
 - Also defined by a **closed loop**.
- **Inside vs. Outside:**
 - **Inside:** The area enclosed by the **exterior ring** but **outside** any interior rings.
 - **Outside:** Everything **beyond** the exterior ring or **inside** the interior rings (the "holes").



Introduction to GIS Data Formats

- **What is a GIS Data Format?**
 - Specifies how **geospatial data** is stored in a file or transmitted between systems.
- **Key Components:**
 - **Geospatial Data:** Describes geographical features.
 - **Metadata:** Includes datum, projection, coordinate system, units, etc.
 - **Attributes:** Additional details like names, population, etc.
 - **Display Information:** Colors, line styles for visualization.



The screenshot shows a GIS application interface. In the background, a map displays several colored polygons (purple, yellow, green, orange) representing different land use or administrative areas. Overlaid on the map is a table window titled "Hydrants.txt Events". The table contains the following data:

ID	XLocation	YLocation	Shape *
16	508878.27	681064.07	Point
18	506543.31	680447.1	Point
19	506877.87	680449.16	Point
20	506958.72	680539.22	Point
21	506758.02	680243.98	Point
22	506587.21	680724.5	Point
23	506783.24	680577.17	Point
24	506419.53	680773.29	Point
25	506430.93	680978.39	Point

Below the table, there are navigation controls (back, forward, first, last) and a status bar indicating "(0 out of 9 Selected)". A red arrow points from the table window to a hydrant symbol on the map.

Metadata: CRS: EPSG:4326 Datum:WGS84

Types of GIS Data Forma

- **Two Main Types:**

- **Raster Formats:**

- Store **bitmapped images** like scanned maps or aerial photos.

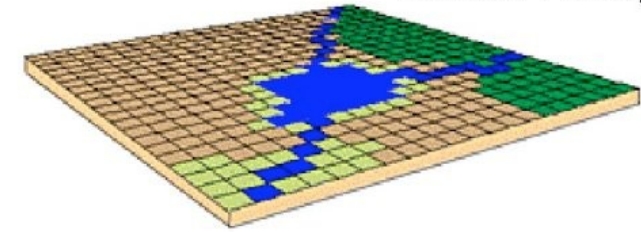
- **Vector Formats:**

- Represent **spatial data** using points, lines, and polygons.

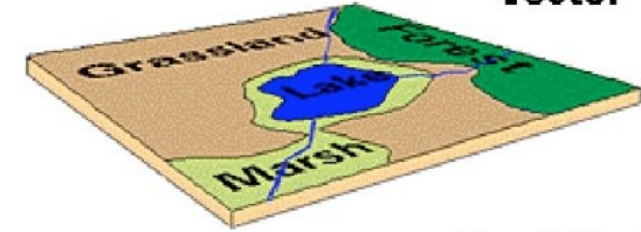
- **Common Usage:**

- **Vector formats** are often preferred for GIS due to smaller file sizes and easier data manipulation.

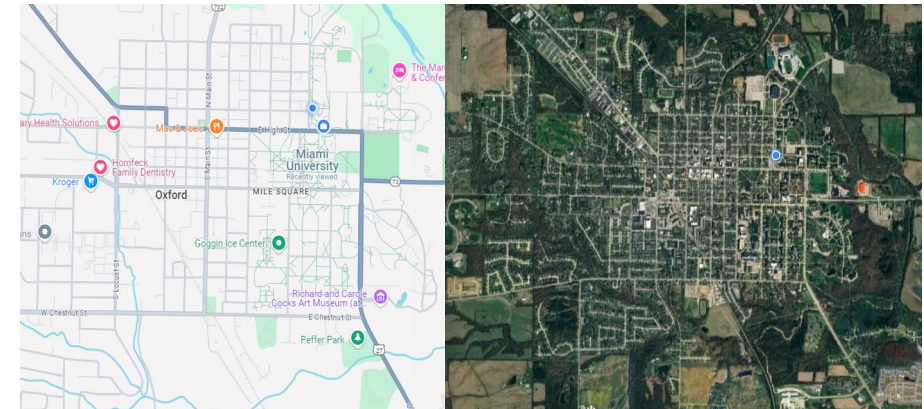
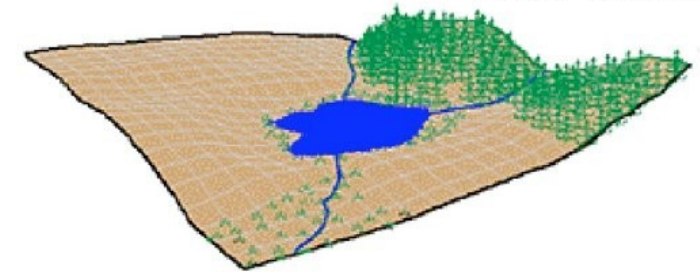
Raster / Image



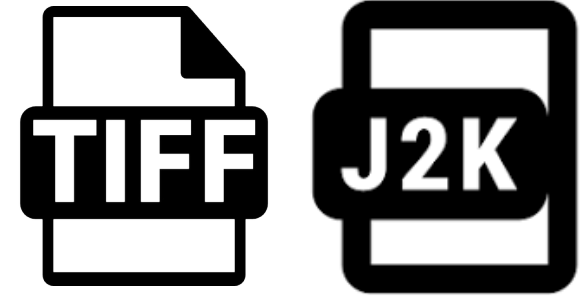
Vector



Real World



Common Raster Formats



- **GeoTIFF:**
 - TIFF with geospatial metadata; widely used in GIS.
- **JPEG 2000 (J2K):**
 - Compressed format for high-quality images with geospatial info.
 - (Not lossy like the JPG format)
- **Digital Raster Graphic (DRG):**
 - Digital scans of paper maps.
- **Digital Elevation Model (DEM):**
 - Captures elevation data for terrain modeling.
- **Remote Sensing Formats:**
 - BIL, BIP, BSQ: Used by on-board hardware in remote sensing.

Vector Formats

Long-Term Storage Vector Formats:

- **Shapefile:** ESRI format with multiple files (e.g., .shp, .shx).
- **Simple Features:** OpenGIS standard for spatial data.
- **TIGER/Line:** Text-based format from the US Census Bureau.
- **Coverage:** Proprietary format used by ESRI's ARC/INFO system.

Vector Formats for Processing & Exchange

- **GeoJSON**: JSON-based format for web GIS.
- **KML**: XML-based format, used in Google Earth.
- **WKT (Well-Known Text)**: Text-based format for individual features.
- **WKB (Well-Known Binary)**: Binary version of WKT.
- **GML**: XML-based open standard for exchanging GIS data.

GeoJSON Overview

- GeoJSON is a format for encoding geographic data structures using JSON.
- Supports various geometry types such as Point, LineString, Polygon, etc.

Basic Structure

- Main Elements:
- **type**: Specifies the GeoJSON object type (FeatureCollection, Feature, Geometry).
- **features**: An array of Feature objects, each representing a spatial entity.
- FeatureCollection is a container for multiple features.

```
{  
  "type": "FeatureCollection",  
  "features": [  
    {  
      // Features go here  
    }  
  ]  
}
```

Feature Object

- Feature contains a geometry and properties:
 - **geometry:**
Defines the shape and location (Point, LineString, etc.).
 - **properties:**
Contains additional data related to the feature (e.g., name).

```
{  
  "type": "Feature",  
  "geometry": {  
    "type": "Point",  
    "coordinates": [-80.191790, 25.761680]  
  },  
  "properties": {  
    "name": "Miami"  
  }  
}
```

Geometry Object

- Geometry types define the shape:
 - **Point**: A single coordinate.
 - **LineString**: A sequence of coordinates forming a line.
 - **Polygon**: A series of closed linear rings.

```
{  
  "type": "Point",  
  "coordinates": [-80.191790, 25.761680]  
}
```

Example GeoJSON

Closing Notes:

- GeoJSON is widely used for representing simple geographical features.
- It's a text-based format, making it easy to share and process.

```
{
  "type": "FeatureCollection",
  "features": [
    {
      "type": "Feature",
      "geometry": {
        "type": "Point",
        "coordinates": [-80.191790, 25.761680]
      },
      "properties": {
        "name": "Miami"
      }
    },
    {
      "type": "Feature",
      "geometry": {
        "type": "Point",
        "coordinates": [-74.006000, 40.712776]
      },
      "properties": {
        "name": "New York City"
      }
    }
  ]
}
```